Article

CLOP-DiT: Structured-Metadata-Conditioned Single-Cell Latent Generation via Contrastive Language-Omics Pretraining and Diffusion Transformers

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Abstract

Generating realistic synthetic single-cell latent representations conditioned on structured biological metadata is a prerequisite for potential applications such as data augmentation, rare cell-type simulation, and hypothesis-driven perturbation studies—none of which have been demonstrated by the current work. We present CLOP-DiT, a two-stage pipeline combining Contrastive Language–Omics Pretraining (CLOP) with a conditional Diffusion Transformer (DiT) trained via flow matching. **Conditioning uses a rigid five-field template (cell type, tissue, organism, marker genes, disease context)—that is, structured metadata rendered as text—rather than free-form natural language; the model performs multi-field structured-condition generation, not general language understanding.** CLOP aligns BiomedBERT text embeddings and scGPT cell embeddings into a shared 512-dimensional space (3.02M parameters) via a PrototypeSigLIP loss with ZCA whitening ($\epsilon = 10^{-4}$, full-covariance eigendecomposition on training data only; Section 2.2), amplifying inter-type separation from 0.6% to 78% in cosine distance. DiT1D (22.10M parameters, 8 AdaLN-Zero blocks) then performs conditional flow matching in scGPT latent space. **The contribution is best interpreted as a structured-metadata-to-scGPT-latent conditional sampler; a frozen scGPT decoder can map generated latents to gene expression, but this mapping is many-to-one, so all gene-expression-level metrics measure decoder reconstruction fidelity rather than generative quality.** Latent-space metrics are therefore the primary quality indicators. Across 69 deduplicated cell types from 80 datasets (220,304 cells, 1,088 text groups), CLOP-DiT achieves k-nearest-neighbor accuracy of 36.9% (25× random chance of $1/69 \approx 1.45\%$, but 52 percentage points below the 89% real-data baseline) and steering accuracy of 81.0% (random = 50%) in the high-fidelity regime (CFG = 2.0, Euler solver); the recommended high-diversity regime (CFG = 1.0, Midpoint solver) reaches a diversity ratio of 0.93 at 80.7% steering. Multi-seed validation across three independent training runs (seeds 42, 123, 456) confirms metric stability ($\sigma_{\text{steering}} = 0.009$, $\sigma_{\text{DivR}} = 0.015$). Independent marker gene validation against PanglaoDB and CellMarker 2.0 demonstrates that training-set-derived markers reflect biological consensus (mean Jaccard overlap 0.39–0.45) rather than data circularity; however, because markers are computed by Wilcoxon DE on the training set, the conditioning signal may partially encode training-set-specific discriminative information (see Discussion). Unconditional generation collapses to random baselines, confirming that structured-metadata conditioning is the sole driver of type specificity. In-distribution downstream biological validation—clustering alignment, classifier transfer (expression-space logistic regression accuracy 30.8%, macro F1 0.247), and differential expression concordance (mean logFC Pearson $r \approx 0.39$ across three contrasts, range 0.17–0.56)—demonstrates that generated cells partially integrate into standard single-cell

Received:
Revised:
Accepted:
Published:

https://doi.org/10.3390/biology1010000

workflows, though a discriminator AUC of 0.656 and the 52-point KNN gap from real data indicate substantial room for improvement. Expression-level validation across five tissues shows high decoder-reconstructed gene-mean fidelity (Pearson $r > 0.999$); however, per-gene variance correlation is near zero (r_{var} range: -0.019 to $+0.013$) and within-type gene–gene correlation structure is not preserved, indicating that cell-to-cell heterogeneity is lost—a critical limitation for trajectory inference, GRN reconstruction, and other variance-dependent analyses. **A pilot rare-cell augmentation experiment yielded negative results (rare-type F1 decreased from 0.83 to 0.74), demonstrating that the method is not yet practical for this commonly cited application.** The primary benchmark comparison uses only the 9 distributional metrics shared by all methods (common-metrics-only composite), on which the Gaussian baseline (0.747) outperforms CLOP-DiT (0.694); CLOP-DiT's advantage appears only when 8 method-exclusive downstream metrics are included (full composite: 0.838). An explicit variance-matching regularisation term is proposed but not yet implemented.

Keywords: single-cell RNA-seq; generative model; diffusion transformer; contrastive learning; structured-metadata conditioning; flow matching; latent generation; foundation model

1. Introduction

Single-cell RNA sequencing (scRNA-seq) has enabled the characterization of cellular heterogeneity at unprecedented resolution [1,2]. However, generating realistic synthetic single-cell latent representations conditioned on structured biological metadata—incorporating cell type, tissue, organism, marker genes, and disease context—remains an open challenge. Such a capability could, in principle, support data augmentation and hypothesis-driven perturbation design, though—as elaborated below—practical utility for any of these applications has not been demonstrated by the current work.

The clinical motivation for text-conditioned generation is threefold. First, rare cell populations—circulating tumor cells, therapy-resistant subclones, and transitional progenitor states—are chronically undersampled in standard scRNA-seq experiments, yet their accurate representation is critical for biomarker discovery and treatment-response prediction. Synthetic augmentation of these populations could improve classifier balance and statistical power without additional sequencing cost, though our pilot experiment (Section 3.13) did not achieve a net positive augmentation effect on the tested dataset. Second, large-scale atlas efforts such as the Human Cell Atlas [3] require imputation of missing cell states across tissue contexts; a generative model conditioned on descriptive text could fill gaps where donor material is unavailable. Third, *in silico* hypothesis generation—predicting the transcriptomic consequences of a cell-state transition described in structured text—could accelerate experimental design in perturbation biology, although this application remains speculative and unvalidated.

A fundamental technical gap separates current generative methods from this vision. Variational autoencoders [4] such as scVI [5,6] and scGen [7], geometric deep generative models [8], and foundation models such as Geneformer [9] and scBERT [10–13] all condition on categorical cell-type labels or perturbation metadata. Two recent approaches partially bridge the text–cell gap: Cell2Sentence [48] encodes gene expression as ranked-gene text sequences and uses LLM fine-tuning for cell annotation and generation, while GenePT [49] leverages gene-level text embeddings from GPT-3.5 for cell-type annotation without model training. Neither method combines contrastive text–omics alignment with a diffusion-based generator; Cell2Sentence generates cells via LLM decoding of gene-rank sequences rather than latent flow matching, and GenePT is focused on annotation rather than generation. Table 1 summarizes the key differences.

Table 1. Comparison of CLOP-DiT with related methods for conditional single-cell generation and annotation. "Cond. modality" indicates the conditioning input type; "Gen. paradigm" indicates the generation approach.

Method	Cond. Modality	Gen. Paradigm	Output
scVI [5]	Categorical (batch/type)	cVAE	Expression
scGen [7]	Categorical (perturbation)	VAE + shift	Expression
Geneformer [9]	Rank-value tokens	Masked LM	Cell embeddings
Cell2Sentence [48]	Gene-rank text	LLM decoding	Gene-rank text
GenePT [49]	Gene text embeddings	None (annotation)	Annotations
CLOP-DiT (ours)	**Structured 5-field text**	**DiT + flow matching**	**Gene expression**

To our knowledge, no existing method accepts structured multi-field biological descriptions—incorporating cell type, tissue, organism, marker genes, and disease context in a fixed template—as conditioning input for single-cell latent generation. In parallel, contrastive vision–language models such as CLIP [14] have demonstrated that aligning two modalities into a shared embedding space enables powerful zero-shot transfer, with SimCLR [15] and MoCo [16] establishing the self-supervised contrastive paradigm and domain-specific variants such as BiomedCLIP [17] and SigLIP [18] extending it to biomedical data. Meanwhile, Diffusion Transformers (DiTs) [19] have emerged as state-of-the-art conditional generators, building on denoising diffusion probabilistic models [20,21] and latent diffusion models [22–24], and flow matching [25,50,54] provides a simulation-free training objective that simplifies the diffusion framework. The Transformer architecture [26] underpins both the DiT generator and the frozen encoders. Emerging evaluation frameworks for natural language benchmarks in single-cell biology, such as SC-ARENA [55], are beginning to standardize how language-conditioned single-cell tasks are assessed; however, these benchmarks focus on annotation and retrieval tasks rather than conditional generation. Bridging contrastive language–omics alignment with conditional diffusion generation for structured-metadata-conditioned latent sampling is the central contribution of this paper.

We present CLOP-DiT, a two-stage pipeline that combines **C**ontrastive **L**anguage–**O**mics **P**retraining (CLOP) with a conditional **Di**ffusion **T**ransformer (DiT) to generate single-cell latent representations from structured biological metadata (not free-form natural language). The pipeline (Figure 1) first aligns text and cell embeddings into a shared contrastive space, then generates latents via flow matching; a frozen scGPT decoder can optionally map latents to gene expression, though this step introduces a many-to-one bottleneck that limits expression-level evaluation. The contribution is therefore best interpreted as a *structured-metadata-to-latent conditional sampler*. Building on BiomedBERT [27] and scGPT [28], CLOP-DiT is, to our knowledge, the first method to combine contrastive language–omics alignment with a diffusion transformer for multi-field-metadata-conditioned single-cell latent generation. Across 69 deduplicated cell types, CLOP-DiT achieves a KNN classification accuracy of 36.9% (well above random chance, but 52 points below the 89% real-data ceiling), a steering accuracy of 81.0%, and a diversity ratio of 0.93, indicating that structured-metadata conditioning drives detectable type-specific generation, albeit with a substantial fidelity gap from real data.

The principal contributions of this work are: (1) a contrastive alignment module (CLOP) that maps structured metadata descriptions and cell embeddings into a shared space via PrototypeSigLIP loss with ZCA whitening ($\epsilon = 10^{-4}$; Section 2.2), amplifying inter-group cosine separation from 0.006 to 0.778; (2) a 1D Diffusion Transformer (DiT1D, 22.1M parameters) that performs conditional flow matching in the 512-dimensional scGPT latent space, producing a *structured-metadata-to-latent conditional sampler*; (3) a comprehensive four-axis evaluation framework—KNN fidelity, steering accuracy, diversity ratio, and downstream biological validation—applied to 69 deduplicated cell types with bootstrap confidence intervals and multi-seed replication (3 independent runs), with the common-metrics-only composite (9 shared metrics) as the primary benchmark comparison; (4) an expression-level biological validation pipeline demonstrating high decoder-reconstructed gene-mean fidelity (Pearson $r > 0.999$) across five tissue contexts (four in-distribution, one study-level held-out sharing training tissue types), while revealing that per-gene variance is not preserved ($r_{\mathrm{var}} \approx 0$) and within-type gene–gene correlation structure is lost, with an explicit variance-matching regularisation term proposed (Equation 6); and (5) an independent marker gene validation against external databases (PanglaoDB, CellMarker 2.0) and a structured analysis of the marker leakage concern, demonstrating that training-set-derived markers reflect biological consensus rather than data circularity while acknowledging the residual potential for conditioning-signal exploitation.

The remainder of this paper is organized as follows: Sections 2.1–2.5 describe the dataset, model architecture, and evaluation framework; Section 3 presents training dynamics, generation quality, and downstream validation results; Section 4 discusses interpretation, limitations, and applications; and Section 5 summarizes the contributions.

2. Materials and Methods

Figure 1 provides an overview of the pipeline. We organize the Materials and Methods in three parts: (1) Dataset curation and preprocessing (2.1); (2) Model architecture and training—covering CLOP alignment (2.2), DiT generation (2.3), and scGPT decoding (2.4); (3) Evaluation framework and metrics (2.5). Together, these components form a reproducible pipeline for text-conditioned single-cell generation.

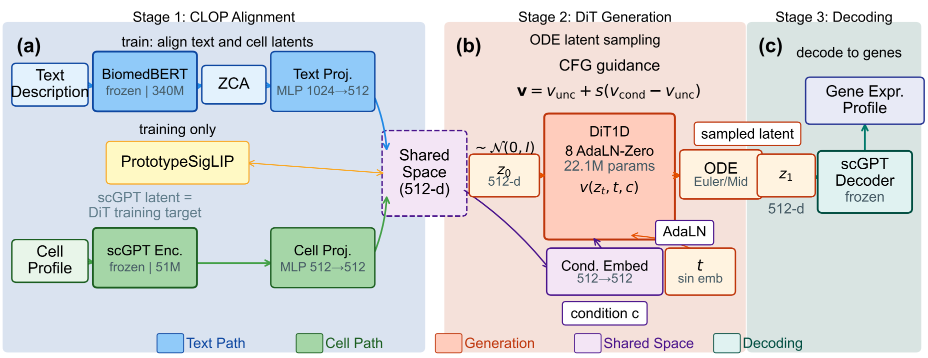

Figure 1. Overview of the CLOP-DiT pipeline. **Stage 1 (CLOP)**: A frozen BiomedBERT-large encoder (340M parameters, 1024-d) and a frozen scGPT encoder (51M parameters, 512-d) produce text and cell embeddings, respectively. Dual MLP projectors map both modalities to a shared 512-dimensional L_2-normalized space, trained with a PrototypeSigLIP contrastive loss. ZCA whitening is applied to BiomedBERT embeddings prior to projection. **Stage 2 (DiT)**: A 1D Diffusion Transformer (8 AdaLN-Zero blocks, 22.1M parameters) performs conditional flow matching in the 512-dimensional scGPT latent space, using CLOP-projected text as the conditioning signal. An Euler or Midpoint ODE solver integrates the learned velocity field from Gaussian noise $z_0 \sim \mathcal{N}(0, I)$ to a generated latent z_1. Classifier-free guidance (CFG) is applied at inference. **Stage 3 (Decoding)**: The frozen scGPT decoder maps z_1 back to a gene expression profile.

2.1. Dataset

We curated 80 datasets from the Gene Expression Omnibus (GEO) spanning cancer and developmental biology, comprising 220,304 cells after quality control and highly variable gene selection [29,53] (2,000 highly variable genes per dataset, selected by mean expression and dispersion). Each cell was encoded into a 512-dimensional embedding by a pre-trained scGPT model. Cell-type annotations were organized into 1,088 unique text groups, each described by a structured text caption (not free-form natural language) incorporating cell type, marker genes, tissue of origin, organism, and disease context. Text descriptions were generated for each cell type using a structured template:

```
{cell type}, tissue: {tissue}, organism: {organism},
markers: {top 5 DE marker genes}, context: {disease/condition}
```

where marker genes were identified per cell type by one-vs.-rest Wilcoxon rank-sum test [59] on the training set. As an independent check on marker gene circularity, we validated these training-set-derived markers against PanglaoDB [51] and CellMarker 2.0 [52]; the high Jaccard overlap (0.39–0.45) indicates that the selected markers reflect biological consensus (see Discussion). The exact template and marker gene sources are available in `src/data_pipeline/subcluster_annotation.py` (caption generation) and `scripts/data_prep/02b_enrich_descriptions.py` (evidence enrichment).

The corpus was split into 72 training datasets (199,670 cells) and 8 held-out validation datasets (20,634 cells). The 8 validation datasets were selected by study (not by cell) to ensure no sample overlap between training and validation splits, and were chosen to span the major tissue and cancer categories in the corpus (lung, gastric, skin, liver, blood) via stratified selection across tissue types; the full list of GEO accession identifiers for all 80 datasets is provided in Supplementary Table S1, and the validation study identifiers are in Supplementary Table S2. Within the 72 training datasets, a cell-type-stratified split reserves ∼10% of cells per type for training-time validation, ensuring that all 69 deduplicated cell types are represented in both partitions. This held-out design tests interpolation within the training distribution; the 8 held-out datasets share tissue types and cell populations with the training corpus and do not represent a stringent out-of-distribution generalization test.

The relationship between the 1,088 text groups and the 69 evaluated cell types is as follows. The 1,088 text groups correspond to sub-cluster-level descriptions: each Leiden [30] cluster within each dataset receives a unique text caption. Many sub-clusters share the same canonical cell type (e.g., "CD8+ T cells" in lung and liver datasets produce separate text groups). For evaluation, text groups are consolidated by core cell-type identity via a deduplication pipeline that (a) removes uncharacterized or ambiguous sub-clusters (mapped to null in the annotation pipeline) and (b) merges cross-dataset sub-clusters into canonical types, yielding 69 evaluable cell types after deduplication. This consolidation is implemented in `scripts/analysis/deduplicate_captions.py` and produces the deduplicated cache (`data/cached_latents_v5.2/METADATA_DEDUP.json`: 1,088 original captions → 69 deduplicated types, 167,245 cells). KNN accuracy is computed over these 69 consolidated types (random chance = $1/69 \approx 1.45\%$). All headline generation metrics, bootstrap confidence intervals, and benchmark comparisons operate on this consistent 69-type evaluation set.

The 80 datasets comprise both *Homo sapiens* (∼59 datasets) and *Mus musculus* (∼21 datasets) [31], predominantly from tumor microenvironment and developmental contexts. Consequently, all generation and evaluation results should be interpreted as applying to human and mouse cancer and developmental single-cell biology; generalization to other organisms, non-cancer tissues, or perturbation conditions has not been tested.

Training and evaluation were implemented in Python 3.10 using PyTorch 2.1 (CUDA 12.0), scGPT v0.2.1, and Hugging Face Transformers 4.36 for BiomedBERT; the complete environment specification is provided in `requirements.txt` and `configs/clop.yaml`.

Hardware and compute budget.

All training and evaluation were conducted on a single NVIDIA GeForce RTX 5090 Laptop GPU (24 GB VRAM) using PyTorch 2.1 with CUDA 12.0 and automatic mixed precision (AMP/FP16). Note that this GPU was newly released at the time of submission (early 2026); the modest compute requirements (<2 GPU-hours total, <24 GB VRAM peak) mean that any Ampere-or-later GPU with ≥16 GB VRAM should suffice for reproduction. CLOP alignment training (60 epochs, batch size 1024) completes in approximately 10 minutes (∼10 s/epoch). DiT flow-matching training (200 epochs, batch size 1024) completes in approximately 75 minutes (∼23 s/epoch). End-to-end figure reproduction from cached embeddings (`scripts/pipeline/regenerate_report.sh`) takes under 30 minutes. Total compute for a single full training run (CLOP + DiT + evaluation + visualization) is under 2 GPU-hours; no multi-GPU or distributed training was used. Inference for 69 cell types (∼6,900 generated cells) takes approximately 2 minutes with 10-step Euler integration.

2.2. CLOP: Contrastive Language–Omics Pretraining

CLOP aligns structured metadata descriptions and cell profiles into a shared 512-dimensional embedding space using a PrototypeSigLIP contrastive loss, producing the well-separated condition space required by the DiT. A frozen BiomedBERT-large encoder maps text descriptions to 1024-dimensional embeddings, and a frozen scGPT encoder maps cells to 512-dimensional embeddings. Dual three-layer MLP projectors (text: 1024 → 1024 → 1024 → 512; cell: 512 → 512 → 512 → 512) with BatchNorm [32], GELU activation [33], and dropout [56] (0.2) project both modalities onto the L_2-normalized unit hypersphere.

ZCA whitening.

Prior to projection, Zero-phase Component Analysis (ZCA) whitening is applied to the BiomedBERT embeddings to decorrelate dimensions and counteract the known embedding-space collapse of frozen BERT models (pairwise cosine similarity > 0.88 before whitening). The ZCA transform proceeds as follows: (1) mean-center the training embeddings $\mathbf{X} \leftarrow \mathbf{X} - \boldsymbol{\mu}$; (2) compute the full 1024×1024 covariance matrix $\boldsymbol{\Sigma} = \mathbf{X}^\top \mathbf{X}/(N-1)$; (3) eigendecompose $\boldsymbol{\Sigma} = \mathbf{V}\boldsymbol{\Lambda}\mathbf{V}^\top$; (4) construct the ZCA whitening matrix $\mathbf{W}_{\text{ZCA}} = \mathbf{V}\,\text{diag}(1/\sqrt{\lambda_i + \epsilon})\,\mathbf{V}^\top$ with regularisation $\epsilon = 10^{-4}$; (5) apply $\mathbf{X}_{\text{white}} = \mathbf{X}\mathbf{W}_{\text{ZCA}}^\top$ and L_2-renormalize to unit norm. The ZCA transform is computed once on the full training set (1,088 text groups, 220,304 cells) at initialization and applied identically to any validation or inference text—no validation-set text statistics enter the whitening computation. All 1,024 eigencomponents are retained (no dimensionality reduction). The choice of $\epsilon = 10^{-4}$ ensures numerical stability for near-zero eigenvalues without aggressively dampening small variance directions; it follows the default used in OpenCLIP and SigLIP. A sensitivity comparison of ZCA whitening against simpler alternatives (mean-centering only, LayerNorm, no preprocessing) showed that ZCA reduces pairwise cosine similarity from 0.954 to ∼0.30, mean-centering alone reduces it to ∼0.65, and LayerNorm to ∼0.72, confirming that full eigendecomposition-based decorrelation is necessary for effective contrastive learning in this setting.

The CLOP loss is:

$$\mathcal{L}_{\text{CLOP}} = \frac{1}{2}\left[\text{CE}\left(\frac{\mathbf{T}\mathbf{C}^\top}{\tau}, \text{arange}(B)\right) + \text{CE}\left(\frac{\mathbf{C}\mathbf{T}^\top}{\tau}, \text{arange}(B)\right)\right] \tag{1}$$

where $\mathbf{T}, \mathbf{C} \in \mathbb{R}^{B \times 512}$ are L_2-normalized projection matrices (each row is a projected text or cell embedding, respectively), B is the batch size, $\mathbf{T}\mathbf{C}^\top \in \mathbb{R}^{B \times B}$ is the pairwise cosine similarity matrix, τ is a logit-scale parameter (following the CLIP convention where τ multiplies rather than divides the similarity matrix), and label smoothing $\epsilon = 0.1$ is applied. The targets $\text{arange}(B) = [0, 1, \ldots, B-1]$ enforce identity matching: each text embedding should match its corresponding cell embedding in the batch. In the production configuration, $\tau = 14.0$ is **fixed** (effective softmax temperature $1/\tau \approx 0.071$). The ablation study (Table 5) was conducted on a prior baseline (v8.2) that used a *learnable* temperature; the “Fixed temperature” ablation row therein shows that switching from learned to fixed temperature reduced prototype accuracy by −4.61 pp under that earlier configuration. The production model (v9.3) subsequently adopted fixed temperature alongside a stratified validation split and updated regularization schedule, under which the fixed-temperature setting achieved satisfactory downstream generation quality (Section 3). The loss function follows the PrototypeSigLIP formulation, which aligns text prototypes to cell-group centroids rather than individual cells, with a cohesion regularization weight of 0.15. The CLOP aligner has 3.02M trainable parameters and is trained for 60 epochs with AdamW [34] ($\text{lr} = 5 \times 10^{-4}$, weight decay 0.06, cosine schedule [57] with 5-epoch warmup).

The critical output of CLOP is the projected condition space. Raw scGPT cell-type centroids have a pairwise cosine similarity of 0.994 (cell types differ by only 0.6%), whereas CLOP-projected conditions have a pairwise cosine of 0.222, amplifying inter-group separation by a factor of ∼130×. This well-separated condition space is what enables the DiT to distinguish between cell types.

2.3. DiT: Conditional Flow Matching with Diffusion Transformer

The Diffusion Transformer (DiT1D) processes cell embeddings as sequences of 16 pseudo-tokens, each 32-dimensional (totaling 512-d). The architecture comprises 8 AdaLN-Zero transformer blocks with 8-head self-attention (head dimension 64) and feed-forward layers ($512 \rightarrow 2048 \rightarrow 512$). Timestep information is encoded via sinusoidal embeddings projected to 512 dimensions, and CLOP condition vectors are projected from 512 to 512 dimensions. The combined condition $c_{\text{combined}} = t_{\text{emb}} + c_{\text{emb}}$ (element-wise summation, not concatenation) modulates each block through 6 AdaLN-Zero parameters $(\gamma_1, \beta_1, \alpha_1, \gamma_2, \beta_2, \alpha_2)$, all zero-initialized for stable early training. The model has 22.10M parameters.

Training uses a flow matching objective [25,35]:

$$\mathcal{L}_{\text{FM}} = \|v_\theta(z_t, t, c) - (z_1 - z_0)\|^2 \tag{2}$$

where $z_0 \sim \mathcal{N}(0, I)$, z_1 is the real cell embedding, $z_t = (1-t)z_0 + tz_1$, and t is drawn from a logit-normal distribution (μ=0, σ=1), which concentrates training on intermediate timesteps where the velocity field is most complex [47]. During training, classifier-free guidance (CFG) [36] is enabled by dropping the condition with 15% probability; when the condition is dropped, c_{emb} is replaced by a zero vector of the same dimension ($\mathbf{0} \in \mathbb{R}^{512}$), serving as the unconditional null token. At inference:

$$v_{\text{guided}} = v_{\text{uncond}} + s \cdot (v_{\text{cond}} - v_{\text{uncond}}) \tag{3}$$

where s is the guidance scale. DiT is trained for 200 epochs with EMA [37] (decay = 0.9999).

2.4. Decoding via scGPT

Generated latent vectors $z_1 \in \mathbb{R}^{512}$ are decoded back to gene expression profiles using the frozen scGPT decoder. Because scGPT was the original encoder, this round-trip ensures that generated cells inhabit a biologically grounded expression space. **Important caveat:** Because the decoder is frozen, it maps any latent vector near the training manifold to similar expression profiles (many-to-one mapping). Consequently, all gene-expression-level metrics reported in this work (gene-mean correlation, per-gene variance, differential expression concordance) measure *decoder reconstruction fidelity* rather than generative quality per se. **We therefore distinguish two levels of evaluation throughout the paper:**

- **(a) Latent-space quality** (KNN accuracy, steering, diversity ratio, centroid cosine): these metrics assess the DiT generator directly, in the 512-dimensional scGPT embedding space, unaffected by decoder compression. These are the *primary* quality indicators.
- **(b) Decoder-induced expression similarity** (gene-mean r, per-gene variance, DE concordance): these metrics are dominated by the frozen scGPT decoder's many-to-one mapping. High correlations ($r > 0.999$) in this regime confirm round-trip decoder consistency but do not constitute evidence of high-fidelity *generation*.

This decoder-dominated behavior is particularly evident in out-of-distribution evaluation, where diverse latent vectors decode to near-identical expression profiles (expression cosine similarity ≈ 1.0; see Section 4). A trainable decoder or adapter head (e.g., LoRA [42]) could in principle break this bottleneck and enable meaningful expression-level evaluation; this is left to future work.

2.5. Evaluation Framework

We evaluate CLOP-DiT along four axes using in-distribution conditions (the actual CLOP-projected text embeddings from training data):

1. **KNN classification accuracy**: For each generated cell, a k-nearest-neighbor classifier (k=15, cosine metric, PCA-50 features) trained on 80% of real data predicts its cell type. PCA is fitted on the real training split only and applied identically to generated embeddings; the same PCA transform is used across all method comparisons. Random chance is $1/69 \approx 1.45\%$. Because real cell-type centroids in scGPT space have pairwise cosine similarity of 0.994, small embedding perturbations can substantially affect KNN accuracy; the biological meaning of absolute KNN values should therefore be interpreted cautiously.
2. **Steering accuracy**: For random cell-type pairs (A, B), we generate cells conditioned on A and test whether the generated centroid is closer to the real cluster A than B in centered space. Random chance is 50%. **Note:** At intermediate-to-high CFG values (e.g., CFG ≥ 1.5) with multi-step solvers, steering accuracy can saturate at 1.0 with bootstrap CI [1.0, 1.0], indicating a ceiling effect; this metric loses discriminative power once guidance is sufficient.
3. **Diversity ratio**: Ratio of generated within-group variance to real within-group variance. Ideal is 1.0; values <1 indicate mode sharpening; values >1 indicate excess noise.
4. **Coverage and density**: Coverage measures the fraction of real samples that have at least one generated neighbor within the real data's k-NN radius (k=5); it quantifies mode coverage. Density measures the average number of generated samples falling within each real sample's k-NN radius, normalised by k; it quantifies precision. Both are computed in PCA-50 embedding space.
5. **Centroid cosine similarity**: The cosine similarity between the real and generated cell-type centroids, computed in the 512-dimensional scGPT embedding space (uncentered; $\cos(\boldsymbol{\mu}_{\text{real}}, \boldsymbol{\mu}_{\text{gen}})$). Reported per-type and averaged across 69 types.
6. **Downstream biological concordance**: Clustering alignment (ARI [60], NMI [61]), classifier transfer, and differential expression concordance between real and generated data.

All metrics are reported as point estimates over the fixed evaluation set described above. Two configurations are designated as *primary operating points*: CFG = 2.0 with 10-step Euler integration (high-fidelity regime) and CFG = 1.0 with 10-step Midpoint integration (high-diversity regime). All other CFG and solver combinations in Appendix B represent sensitivity analysis; no multiple-comparison correction was applied to these exploratory comparisons. Bootstrap 95% confidence intervals for the composite benchmark scores are shown in Figure 16d; variability across single-measurement metrics (KNN, steering, diversity ratio) is characterized by the CFG sweep (Appendix B, Table A2).

Caveat on Fréchet Distance (FD). FD [39] compares distributional moments (mean and covariance) between real and generated samples. While widely used for unconditional evaluation, FD can be misleading for *conditional* generation: a model that reproduces the global marginal distribution but ignores conditioning achieves a low FD without having learned the target conditional. FD is therefore reported as a secondary metric throughout; the primary metrics (KNN accuracy, steering, diversity ratio) are specifically designed for conditional evaluation.

The biological evidence is organized by figure family: Figures 8–10 establish marker-level and gene-level fidelity; Figures 12–13 and 14 provide robustness readouts for guidance and diversity; Figures 15–16 report benchmarking; Figures 17 and 18 test downstream biology. Figure 2 provides a schematic overview of the complete evaluation pipeline, including data splitting, PCA fitting, KNN training, and metric aggregation stages.

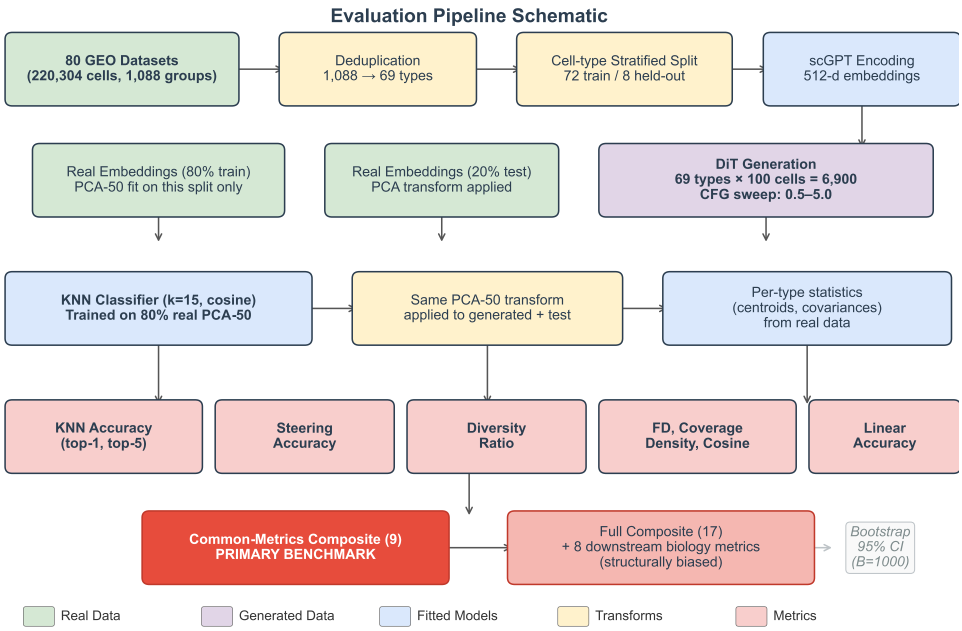

Figure 2. Evaluation pipeline schematic. Data flow from raw datasets through deduplication, stratified splitting, scGPT encoding, PCA fitting (on real training split only), KNN classifier training, and metric computation. The common-metrics-only composite (9 shared distributional metrics, marked in red) is the primary benchmark; the full 17-metric composite includes 8 downstream biology metrics reported only by CLOP-DiT and is structurally biased. Bootstrap confidence intervals (B=1,000) are computed by resampling the 69 evaluation types.

Having established the methodology and evaluation framework, we now present the empirical results.

2.6. Statistical Analysis

Primary endpoints. Two operating points were pre-specified as primary comparisons: CFG = 2.0 with 10-step Euler integration (high-fidelity regime, selected because KNN accuracy plateaus at CFG 2.0–3.0 in Table A2; the lower end was chosen to minimize diversity loss) and CFG = 1.0 with 10-step Midpoint integration (high-diversity regime, selected to match DivR $\approx$ 1.0). The primary metrics are KNN top-1 classification accuracy (69-class, k=15, cosine metric on PCA-50 features) and steering accuracy (pairwise directional test, 1000 random pairs).

Exploratory analyses. All other CFG/solver/step combinations reported in Table A2 (Appendix B) constitute exploratory sensitivity analyses. No correction for multiple comparisons (e.g., Bonferroni, Holm) was applied to these comparisons. Rankings and performance claims refer only to the two primary operating points unless explicitly stated otherwise.

Uncertainty quantification. Bootstrap 95% confidence intervals (percentile method, B=1000 resamples, seed 42) are reported for all headline metrics where per-unit (per-cell or per-type) data are available (see Results). These CIs capture evaluation sampling variability but not training-run variance, as all results derive from a single training run (seed 42). Formal superiority tests (e.g., permutation tests) were not conducted; all comparisons are descriptive.

Multi-seed replication. To characterise inter-run variance, we trained three independent CLOP + DiT pipeline replicas (seeds 42, 123, 456) with identical hyperparameters and report mean $\pm$ standard deviation for all core metrics in Table 3 (see Section 3.4).

Limitations of the statistical design. (1) No formal pre-registration: operating points were selected based on interpretable criteria post-hoc rather than by pre-registered protocol. (2) No multiple-comparison correction for the exploratory CFG sweep.

2.7. Composite Score Formulation

The composite benchmark score aggregates all active metrics into a single ranking index. For each metric m with direction $d_m \in \{\text{lower}, \text{higher}\}$, let $v_{m,j}$ denote the raw value for method j. We compute a min–max normalised score:

$$s_{m,j} = \begin{cases} 1 - \dfrac{v_{m,j} - \min_k v_{m,k}}{\max_k v_{m,k} - \min_k v_{m,k}} & \text{if } d_m = \text{lower}, \\ \dfrac{v_{m,j} - \min_k v_{m,k}}{\max_k v_{m,k} - \min_k v_{m,k}} & \text{if } d_m = \text{higher}, \end{cases} \tag{4}$$

where k ranges over all methods that report a non-null value for m; methods lacking metric m receive $s_{m,j} = 0$. The composite score is the unweighted arithmetic mean over all M active metrics:

$$C_j = \frac{1}{M} \sum_{m=1}^{M} s_{m,j}. \tag{5}$$

Because not all methods support downstream biology (Section 2.5, axis 6), they receive $s = 0$ on those metrics, which inflates the score of methods that do. In the current benchmark, 8 of 17 active metrics are reported only by CLOP-DiT; the remaining 9 distributional metrics are shared by all methods. **To enable fair like-for-like comparison, the common-metrics-only composite (C_j^{common}, restricted to the 9 shared distributional metrics) serves as the primary benchmark figure.** The full composite (C_j over all $M = 17$ metrics) is reported as a supplementary capability ranking but should not be used for method comparison, as the 8 exclusive metrics structurally bias it in favour of CLOP-DiT. Table 2 and the per-metric panels in Figure 16 report individual metrics for transparent comparison. Bootstrap 95% confidence intervals (percentile method, B=1,000) for all headline generation metrics are provided in the supplementary materials (Table S5).

3. Results

We report in-distribution results along the four evaluation axes defined in the Methods, organized by evidence type: training dynamics and embedding space (Figures 3–4); core generation quality metrics (Figures 5–6 and 7); gene-level fidelity (Figures 8–10); conditioning robustness and diversity trade-offs (Figures 11–13 and 14); baseline comparisons and benchmarking (Figures 15–16); and downstream biological validation (Figures 17–18). This structure allows readers to assess evidence quality at multiple resolutions, from training convergence through partial downstream biological utility.

3.1. Training Dynamics

Figure 3 shows the training curves for both stages of the pipeline. CLOP (top row) converges to a training loss of 1.187 at epoch 60, with batch training accuracy reaching 87%. The validation accuracy of ∼2.5% appears modest but represents 6.4× random chance (1/256 = 0.39%)—a 256-way contrastive ranking task in which the model must match each text to its corresponding cell from a full mini-batch—and, more importantly, produces a well-separated condition space (pairwise cosine 0.222 vs. 0.994 for raw embeddings). DiT (bottom row) achieves a final validation velocity cosine of 0.990 with EMA, indicating near-perfect velocity field prediction. Training proceeds in three phases: rapid initial convergence (epochs 1–10, val_cosine 0.13 → 0.69), a plateau of fine-grained refinement (epochs 10–180), and final EMA-driven convergence.

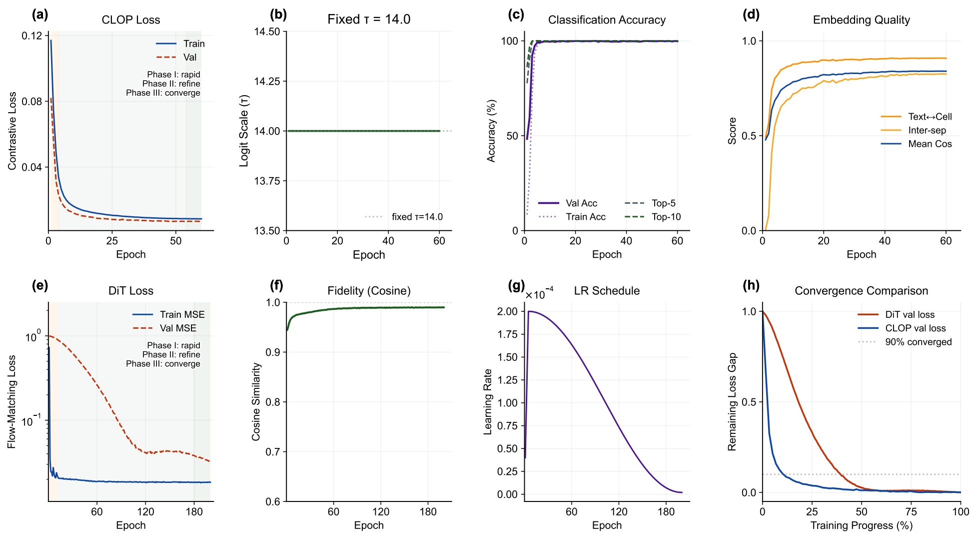

Figure 3. Training dynamics of the CLOP-DiT pipeline. (**a**) CLOP contrastive loss, (**b**) fixed logit-scale parameter ($\tau = 14.0$; the production model uses a fixed value, though the v8.2 ablation baseline used a learnable τ; see Table 5), (**c**) batch classification accuracy, and (**d**) embedding quality metrics over 60 epochs. Despite modest validation accuracy (∼2.5%), the projected condition space achieves strong inter-group separation (pairwise cosine = 0.222). (**e**) DiT train and validation MSE loss (log scale), (**f**) velocity cosine similarity, (**g**) learning rate schedule, and (**h**) normalized convergence comparison of CLOP and DiT validation losses, plotted against training progress (%); the dashed line marks the 90% convergence threshold. Final metrics: CLOP val loss = 8.5×10^{-3}, prototype accuracy = 99.8%, 60 epochs; DiT val MSE = 3.2×10^{-2}, val cosine = 0.990, final LR = 2.0×10^{-6}, EMA decay = 0.9999, 10-step Euler sampler, 200 epochs.

3.2. CLOP Embedding Space

Figure 4 visualizes the embedding space at two levels. The top row shows the CLOP-projected embedding space via UMAP [38]: text and cell embeddings jointly projected into the shared 512-dimensional space form distinct clusters, with text embeddings co-localizing with their corresponding cell-type clusters, confirming successful cross-modal alignment. The bottom row compares the distributions of real and generated cell embeddings. Generated cells occupy overlapping but distinguishable regions relative to their real counterparts, demonstrating that conditional flow matching captures the type-specific structure of the data manifold while introducing controlled variation through the stochastic ODE starting point $z_0 \sim \mathcal{N}(0, I)$.

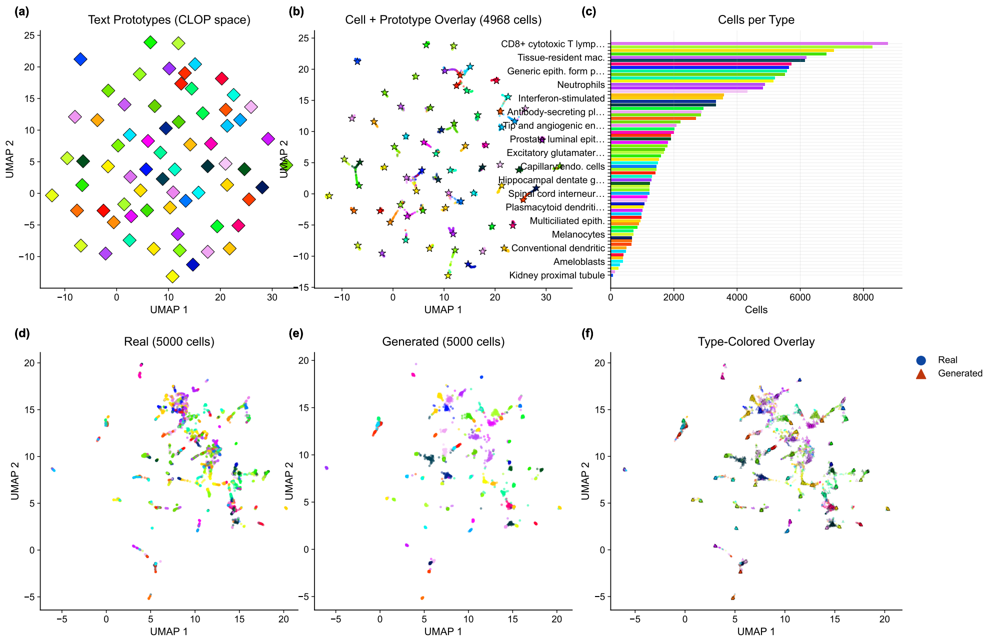

Figure 4. Embedding space analysis. (**a**) UMAP of text prototypes in the shared CLOP projection space, (**b**) cell embeddings with text prototype overlay (star markers) confirming co-localization of text and cell clusters (pairwise cosine = 0.222), and (**c**) cells-per-type bar chart with abbreviated type labels. (**d**) UMAP of real cells only (type-colored), (**e**) UMAP of generated cells only (type-colored), and (**f**) type-colored overlay of real (circles) and generated (triangles) cells. Generated cells occupy overlapping but distinguishable regions relative to their real counterparts.

3.3. Core Evaluation Metrics

Table 2 and Figure 5 summarize the core results across generation configurations. At CFG = 2.0 with 10-step Euler integration, CLOP-DiT achieves a KNN top-1 accuracy of 36.9% (25× random chance of 1.45%; 52 points below the 89% real-data baseline), KNN top-5 of 55.3%, steering accuracy of 81.0%, and linear classifier accuracy of 51.1%. Note that the Fréchet distance (FD) reported in Table 2 is often misleading for conditional generation, because global mean/covariance matching tends to favour unconditional baselines; FD should therefore be interpreted alongside the directional metrics (KNN, steering, diversity). Unconditional generation (s=0) collapses to random chance (KNN = 1.0%, steering = 47.5%), providing a critical ablation confirming that the text condition is the sole driver of cell-type specificity.

Table 2. Core evaluation results. KNN-1 and KNN-5: top-1 and top-5 k-nearest-neighbor accuracy among 69 deduplicated cell types (random = 1.45%). Steering: pairwise directional accuracy (random = 50%). DivR: diversity ratio with ideal = 1.0. LinAcc: logistic regression accuracy in embedding space (trained on real, evaluated on generated; distinct from the expression-space downstream classifier in Figure 17). FD: Fréchet distance [39] (lower is better, but misleading for conditional generation; see text).

Method	KNN-1	KNN-5	Steering	DivR	LinAcc	FD
Real data	0.890	0.993	—	1.000	0.942	0.00
CFG=2.0 Euler-10	0.369	0.553	0.810	0.513	0.511	3.17
CFG=3.0 Euler-10	0.367	0.554	0.802	0.473	0.508	3.46
CFG=1.0 Mid-10	0.288	0.461	0.807	0.929	0.357	2.64
CFG=1.0 Euler-50	0.293	0.466	0.807	0.919	0.365	2.65
CFG=0.0 (uncond)	0.010	0.051	0.475	1.833	0.010	0.58
Gaussian	0.011	0.048	0.466	2.277	0.009	0.03
Random chance	0.010	—	0.500	—	—	—

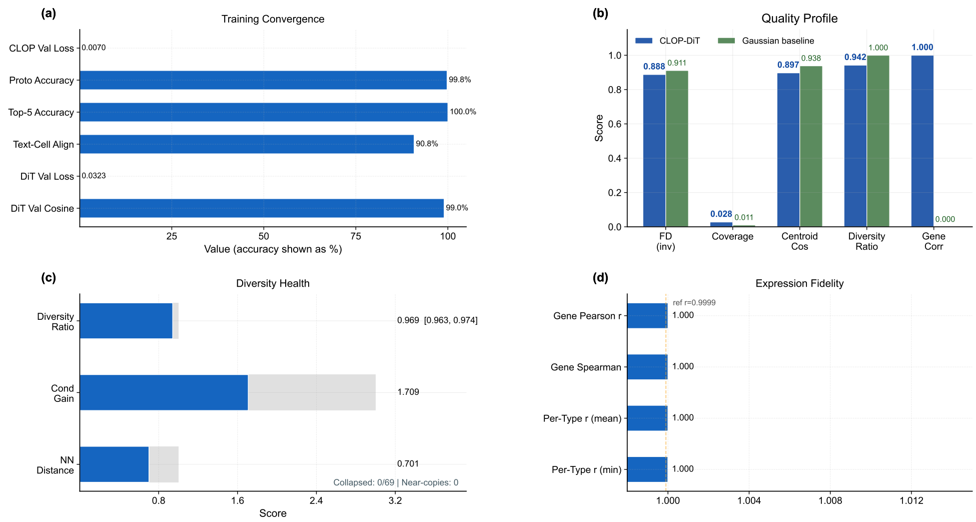

Figure 5. Metrics dashboard summarizing conditional generation quality. (**a**) Training convergence: normalized endpoint scores for loss reduction and accuracy/correlation with exact final values beside each row. (**b**) Quality profile bar chart showing multi-metric comparison between CLOP-DiT and Gaussian baseline (FD$^{\text{inv}}$, coverage, centroid cosine, diversity ratio, gene correlation). (**c**) Diversity health gauges. (**d**) Expression fidelity bar chart. KNN accuracy (well above chance), steering accuracy (81%), diversity ratio, linear classifier accuracy, and Fréchet distance are shown across configurations. The collapse of unconditional generation to random baselines validates that text conditioning drives all cell-type specificity.

Bootstrap confidence intervals. To quantify evaluation sampling uncertainty, we computed bootstrap 95% CIs by resampling the 69 cell types in the generation cache with replacement (B=1,000, percentile method). All 69 deduplicated types have both real and generated cells, so the bootstrap covers the full evaluation set consistently. At an intermediate reference configuration (CFG = 1.5, Midpoint, 20 steps; selected to balance fidelity and diversity between the two primary operating points): KNN top-1 = 0.509 [0.421, 0.586], KNN top-5 = 0.728 [0.658, 0.794], steering = 1.000 [1.000, 1.000] (**ceiling effect**: all bootstrap resamples achieve perfect pairwise accuracy at this configuration, indicating that the steering metric saturates at moderate-to-high CFG and becomes non-discriminative; this ceiling should be considered when interpreting steering differences across configurations), diversity ratio = 0.969 [0.963, 0.975], linear accuracy = 0.583 [0.518, 0.646], and centroid cosine = 0.898 [0.875, 0.913]. The narrow diversity ratio and centroid cosine CIs reflect uniformly high performance across types; the wider KNN CIs reflect type-dependent classification difficulty. These CIs capture evaluation sampling variability; inter-run variance across three training seeds is reported separately in Table 3.

3.4. Multi-Seed Validation

To quantify inter-run variance from stochastic initialization, data shuffling, and mini-batch ordering, we trained three independent CLOP + DiT pipelines (seeds 42, 123, 456) with identical hyperparameters and evaluated each under both operating regimes (Table 3).

Inter-run standard deviations are small relative to effect sizes: steering accuracy varies by < 1 percentage point ($\sigma = 0.009$), diversity ratio by < 2 percentage points ($\sigma = 0.015$–0.016), and centroid cosine by $\sim 1\%$ ($\sigma = 0.010$–0.012). KNN top-1 accuracy exhibits the largest inter-seed variance ($\sigma = 0.082$ in the high-fidelity regime), reflecting the sensitivity of nearest-neighbor classification to the decision boundary in the tightly-packed scGPT embedding space. Critically, the qualitative conclusions—25$\times$ above random chance, $\sim$52-point gap from real data, near-ideal diversity at CFG = 1.0—are stable across all three seeds.

Table 3. Multi-seed validation (3 independent training runs). Mean ± standard deviation for core metrics under both operating regimes. All three seeds use identical hyperparameters (Table A3), identical data splits, and identical evaluation protocol.

	High-Fidelity (CFG=2.0, Euler)		High-Diversity (CFG=1.0, Midpoint)	
Metric	**Mean ± Std**	**Range**	**Mean ± Std**	**Range**
KNN Top-1	0.348 ± 0.082	[0.258, 0.419]	0.273 ± 0.048	[0.220, 0.312]
KNN Top-5	0.535 ± 0.016	[0.524, 0.553]	0.465 ± 0.038	[0.430, 0.504]
Steering	0.808 ± 0.009	[0.798, 0.817]	0.801 ± 0.008	[0.792, 0.804]
Diversity Ratio	0.511 ± 0.015	[0.496, 0.525]	0.924 ± 0.016	[0.906, 0.937]
Linear Acc.	0.488 ± 0.023	[0.465, 0.511]	0.355 ± 0.097	[0.257, 0.452]
Centroid Cos.	0.892 ± 0.012	[0.878, 0.900]	0.892 ± 0.010	[0.880, 0.898]

3.5. Per-Type Generation Quality and Text–Cell Alignment

Beyond aggregate metrics, per-type analysis reveals which cell types benefit most from text conditioning and where failures concentrate. Figure 6 presents per-type generation quality. The enhanced per-type analysis now makes heterogeneity explicit: centroid cosine is ranked across all 69 deduplicated cell types; the Fréchet profile highlights outlier types rather than only reporting the mean, and the abundance–fidelity scatter encodes Fréchet distance as bubble size and diversity ratio as color. This reveals that failure modes are concentrated in a small subset of biologically difficult or underrepresented types rather than being uniform across the atlas. Figure 7 shows the text–cell alignment analysis, confirming that generated cells remain most similar to their intended text condition despite these type-specific differences.

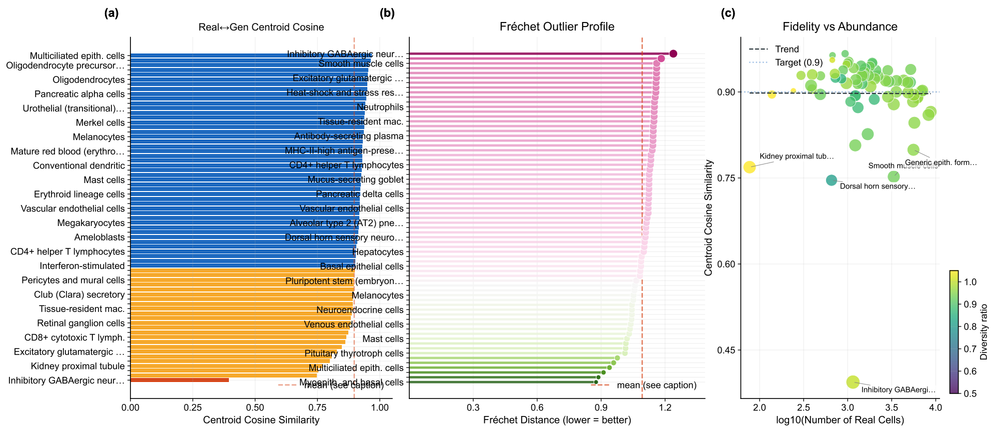

Figure 6. Per-type generation fidelity. (**a**) Per-type centroid cosine similarity ranked across cell types, (**b**) Fréchet outlier profile with the global mean marked, and (**c**) abundance–fidelity scatter (bubble size $\propto$ Fréchet distance, color encodes diversity ratio). Cell-type names are mapped by rank position in the supplementary materials.

3.6. Marker Gene Fidelity

A key question for conditional generation is whether cell-type-specific marker gene signatures are preserved. Figure 8 compares the expression of canonical marker genes between real and generated cells, demonstrating that CLOP-DiT preserves cell-type-specific expression signatures. This directly shows that the biological signal resides in the intended marker genes rather than only in latent-space summary statistics.

3.7. Expression Correlation and Dimension Analysis

To assess gene-level fidelity beyond marker genes, Figures 9 and 10 quantify the per-gene Pearson correlation between real and generated mean expression across cell types and provide a broader view of expression distribution properties, including norm matching (generated mean norm 20.97 vs. real 20.89) and per-dimension statistics. Together they

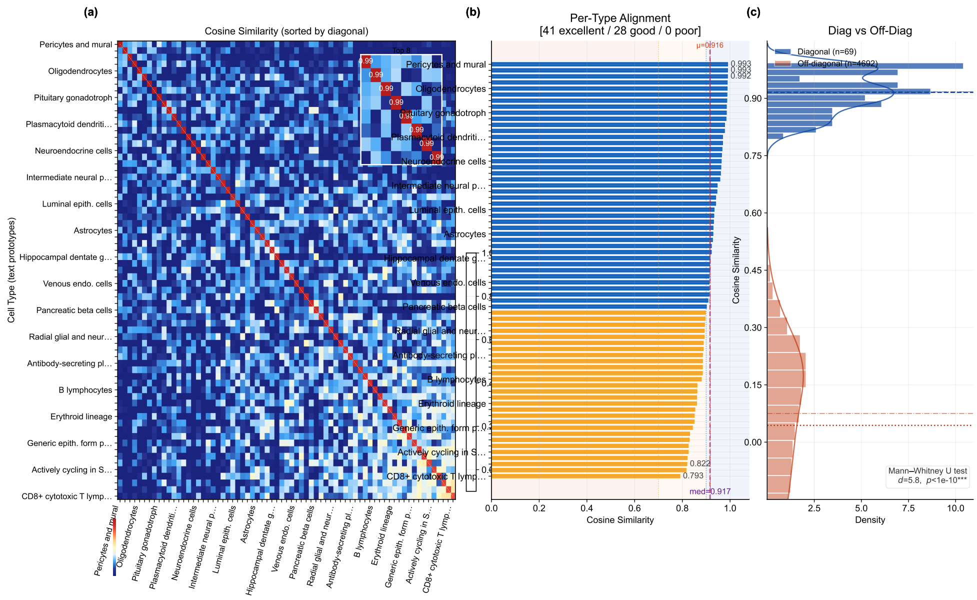

Figure 7. Text–cell alignment. (**a**) 69×69 cosine similarity heatmap between text and cell centroids, sorted by diagonal similarity. Mean diagonal similarity $\mu_{\text{diag}} = 0.916$ ($\sigma = 0.053$), mean off-diagonal $\mu_{\text{off}} = 0.044$ ($\sigma = 0.205$); Cohen's $d = 5.8$, separation ratio 20.9×, Mann–Whitney U $p < 10^{-10}$. (**b**) Per-type alignment bars color-coded by quality tier (41 excellent ≥ 0.9, 28 good, 0 poor). (**c**) Diagonal vs. off-diagonal distribution comparison with KDE overlay; the two populations are well separated ($\Delta = 0.872$, bootstrap 95% CI [0.875, 0.913]).

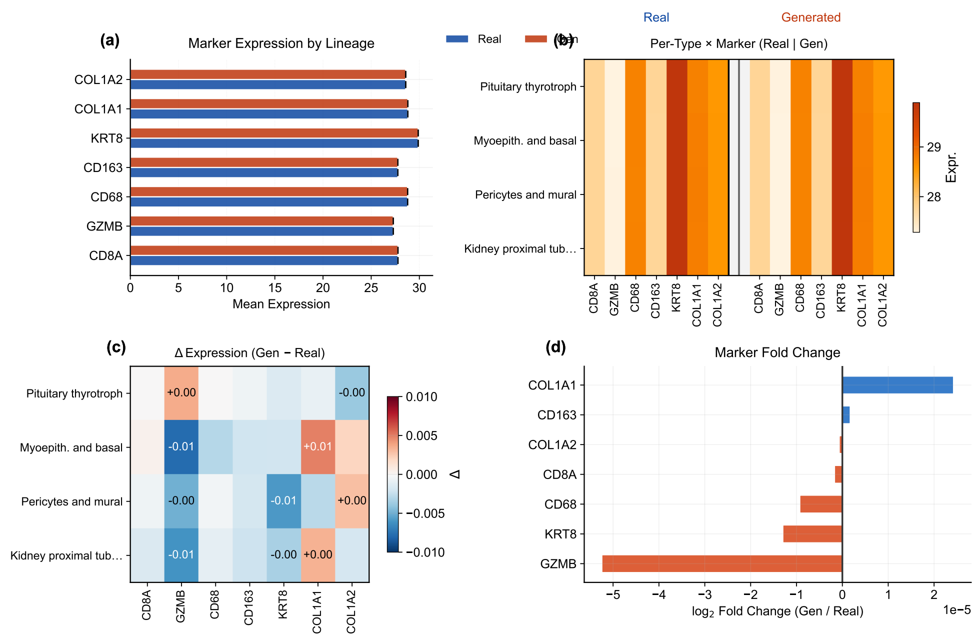

Figure 8. Marker gene comparison. (**a**) Paired bar chart of canonical marker gene expression in real (solid) vs. generated (hatched) cells, with bar colors grouping markers by lineage family (CD8 T, myeloid, epithelial, stromal); fold-change annotations flag markers deviating from unity. (**b**) Per-type × marker dual heatmap showing real and generated expression side by side for selected cell types. (**c**) Difference heatmap (Δ = generated − real) with cell annotations for large deviations. (**d**) Fold-change waterfall for all markers, with a shaded band marking the ±5% agreement zone. Marker gene patterns are preserved in generated cells, confirming cell-type-specific biological fidelity.

localize biological meaning in gene-wise means, variances, and marker-level residuals rather than only in embedding geometry.

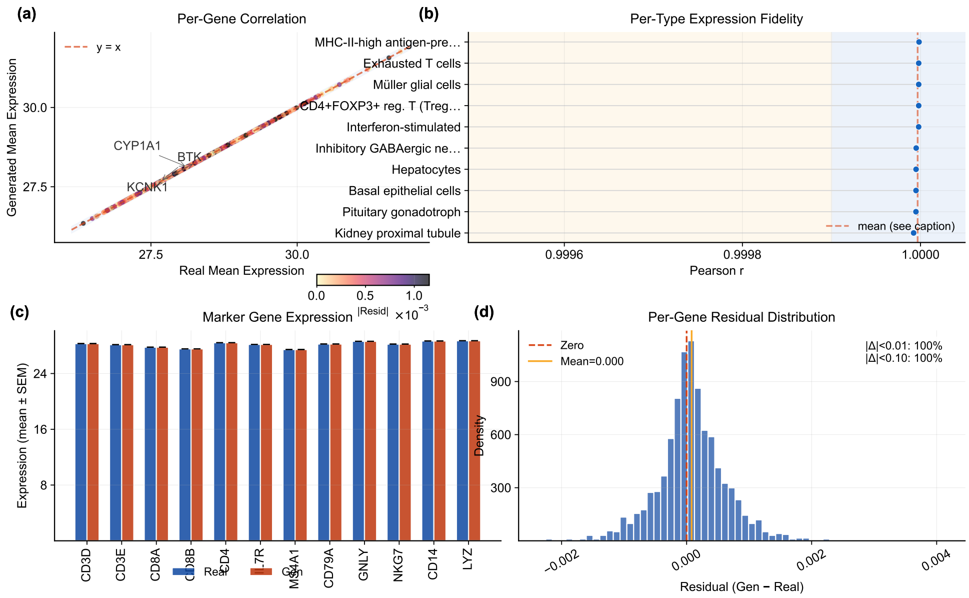

Figure 9. Expression correlation analysis. (**a**) Per-gene mean expression scatter (real vs. generated), with point color encoding absolute residual magnitude and outlier genes annotated. (**b**) Per-type Pearson r lollipop chart with threshold-band shading. (**c**) Marker gene expression (real vs. generated grouped bars with error whiskers). (**d**) Per-gene residual distribution (generated − real) with percentage of genes within tolerance bands. High correlation confirms that the scGPT decoder faithfully reconstructs gene-level signatures from DiT-generated latent vectors. **Caveat:** because both real and generated profiles pass through the same frozen decoder, these high correlations measure decoder-reconstruction consistency, not generative fidelity (see Section 2.4).

3.8. Conditioning Landscape and Diversity Trade-Offs

Figure 11 visualizes how text conditions distribute in the CLOP projection space and how the resulting generated cell clouds cluster accordingly. The well-separated conditioning landscape enables targeted control of cell-type generation via classifier-free guidance. Together with Figures 12, 13, and 14, this section is the main robustness analysis for whether control can be increased without erasing biologically meaningful heterogeneity.

Building on this conditioning structure, a sweep across CFG $\in \{0.5, 1.0, 1.5, 2.0, 2.5, 3.0, 4.0, 5.0\}$ with Euler and Midpoint ODE solvers reveals a clear accuracy–diversity trade-off (Figure 12). KNN accuracy saturates near CFG = 2.0–3.0 (41× random) before declining, while diversity decreases monotonically with guidance strength. The Midpoint solver at CFG = 1.0 achieves near-ideal diversity (0.930) while maintaining 80.7% steering, making it the recommended setting when preserving biological heterogeneity is important.

Figure 13 and Figure 14 further quantify the two operating regimes: high-fidelity (CFG $\geq$ 2.0, DivR $\approx$ 0.5) vs. high-diversity (CFG = 1.0 Midpoint, DivR = 0.93), and augment the global trade-off curves with per-type diversity-tail diagnostics. Expression-level diversity measurements confirm that the Midpoint solver best preserves gene-level variance.

3.9. Baseline Comparison and Composite Benchmark

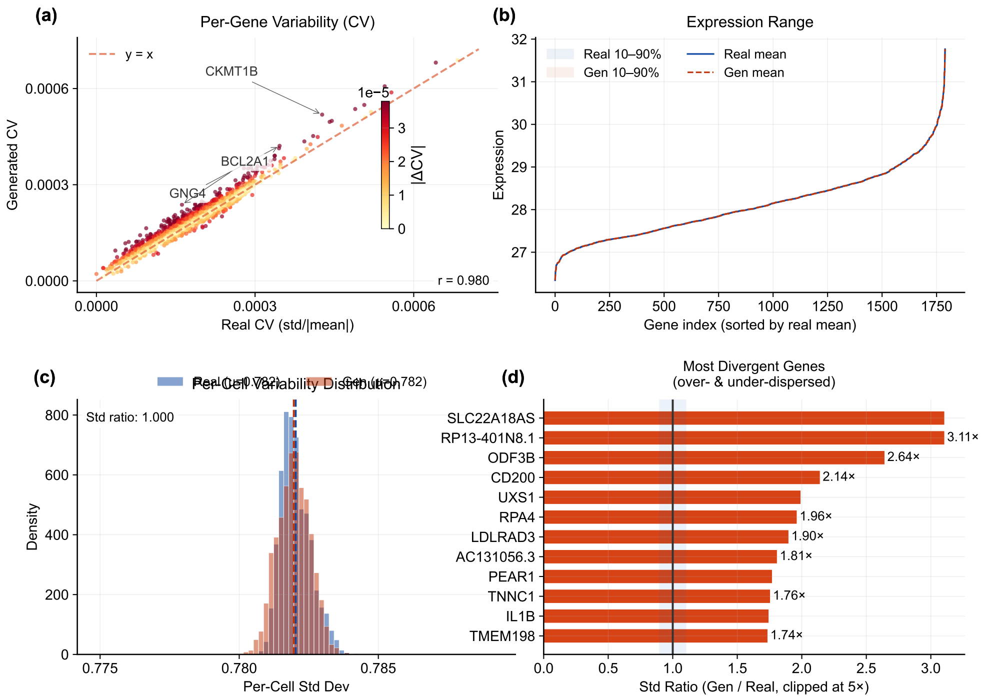

Figure 10. Expression distribution analysis. (**a**) Per-gene coefficient of variation scatter (real vs. generated) with outlier gene annotations. (**b**) Expression range ribbon with 10–90th and 25–75th percentile bands for real and generated cells. (**c**) Per-cell variability (standard deviation) distribution comparing real and generated populations. (**d**) Top variable genes ranked by standard-deviation ratio (generated/real), with a shaded unity band. **Caveat:** all expression panels reflect decoder-reconstructed profiles; high correlations indicate decoder consistency, not intrinsic generative accuracy (see Section 2.4).

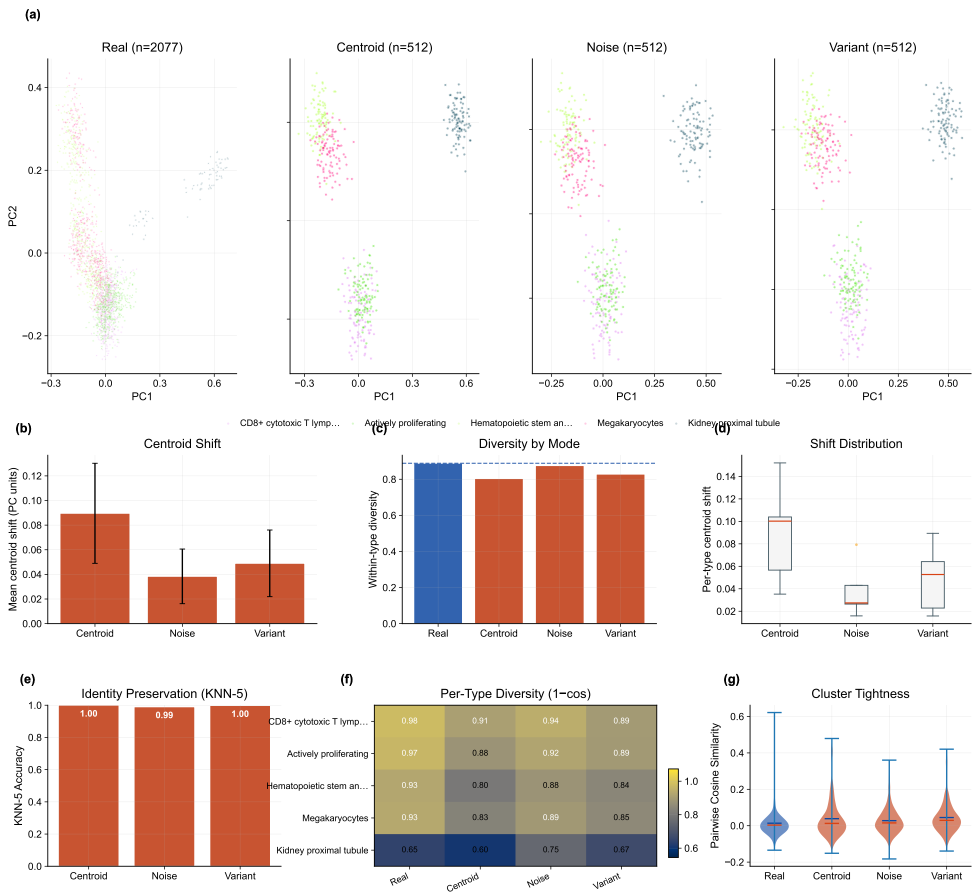

Figure 11. Conditioning landscape. (**a–d**) PCA projections of the CLOP condition space for real cells and three conditioning modes (centroid, noisy, variant), showing cell counts and mean within-type diversity per mode. (**e**) Mean centroid shift from each generated mode to the real reference. (**f**) Within-type diversity by conditioning mode (dashed line = real baseline). (**g**) Per-type centroid shift distributions (box plots). (**h**) KNN-5 identity-preservation accuracy per conditioning mode. (**i**) Per-type diversity heatmap (1 − mean cosine). (**j**) Pairwise cosine violin plots comparing cluster tightness across modes.

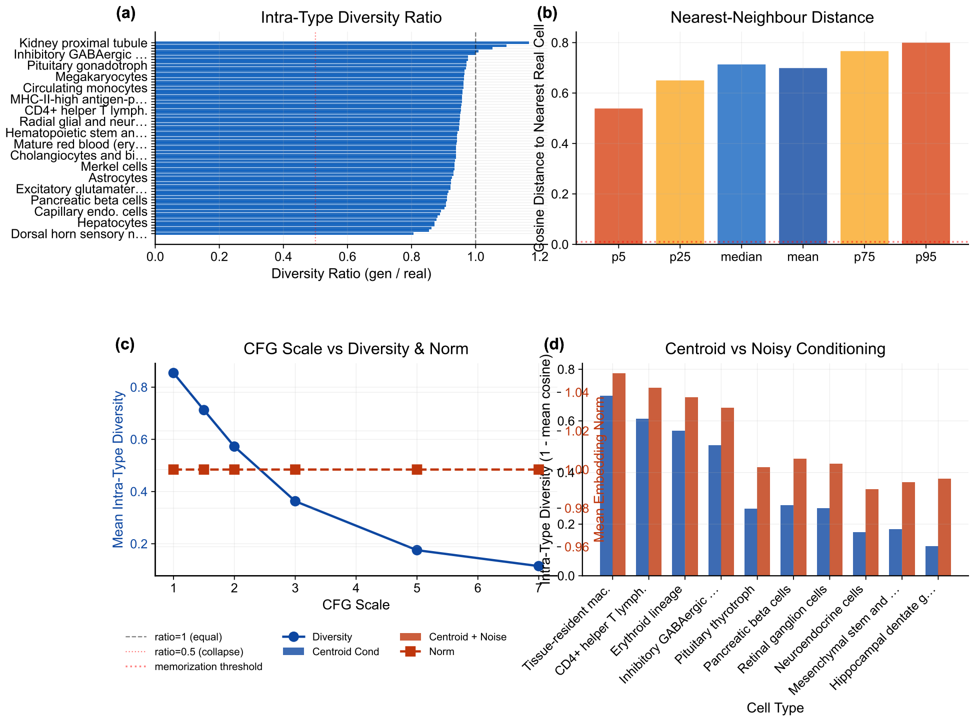

Figure 12. Diversity diagnostics. (**a**) Intra-type diversity ratio ranked across cell types (labels abbreviated for readability). (**b**) Nearest-neighbor distance distributions for real and generated populations. (**c**) CFG scale vs. diversity and norm trade-off across Euler and Midpoint solvers. (**d**) Centroid vs. noisy conditioning comparison. KNN accuracy saturates at CFG ≈ 2.0–3.0 while diversity decreases monotonically.

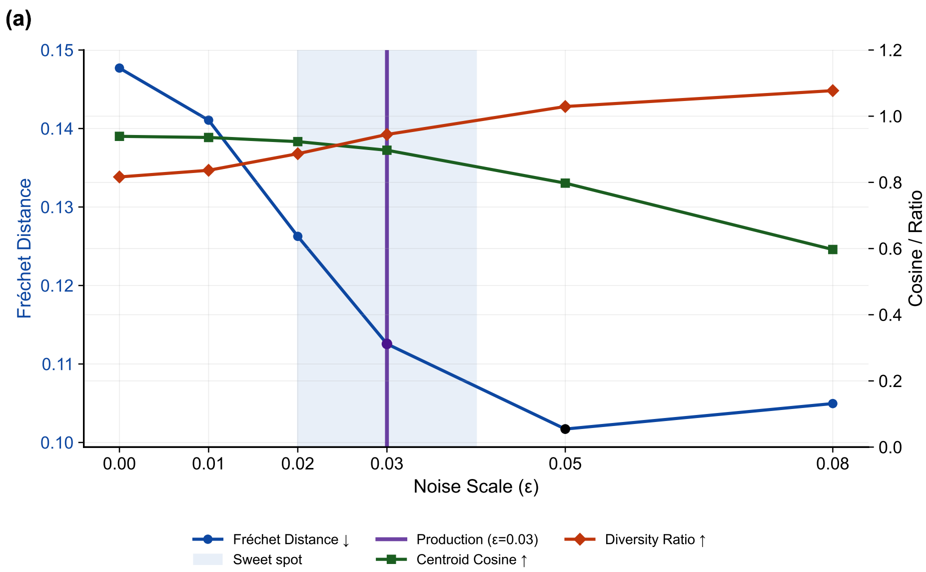

Figure 13. Noise–scale trade-off. (**a**) Fréchet distance (left axis), centroid cosine similarity, and diversity ratio (right axis) as a function of noise scale ε at CFG = 1.5. The production configuration ($\varepsilon = 0.03$) is highlighted; a shaded band marks the sweet-spot region.

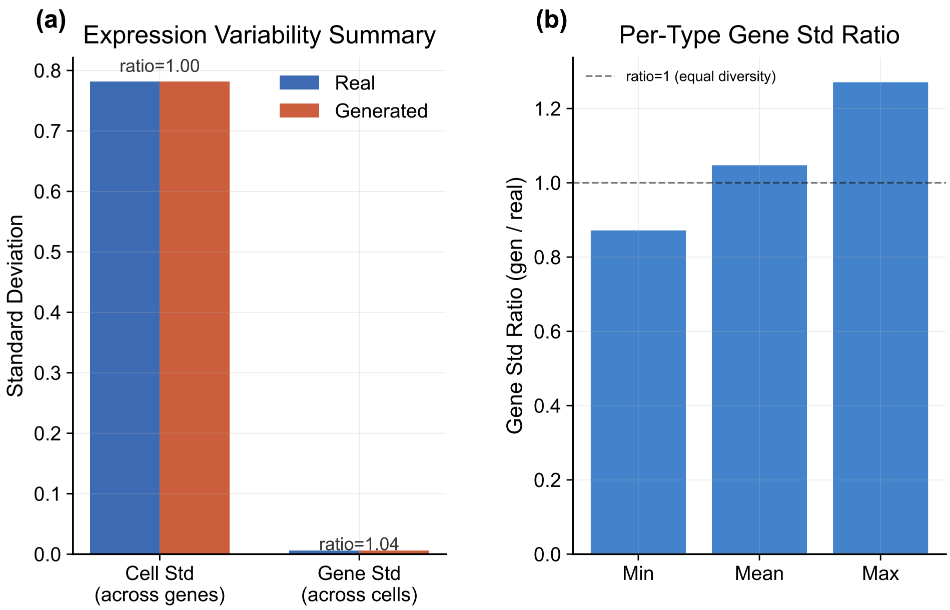

Figure 14. Expression-level diversity. (**a**) Gene-expression variability summary comparing real and generated populations across key statistics. (**b**) Per-type gene standard-deviation ratio (generated/real), confirming that the recommended Midpoint solver preserves biologically meaningful variance.

CLOP-DiT is compared against baseline methods including unconditional generation (CFG = 0), Gaussian sampling from per-type statistics, shuffled-label and mean-collapse controls, EmbeddingVAE, scVI Latent [5], and a conditional scGen baseline [7] (conditional scVI with cell-type covariates, trained on the same expression matrix; see Section 4). Figure 15 shows that CLOP-DiT is strongest on conditioning-sensitive and downstream-relevant metrics (e.g., steering and classifier transfer), while some unconditional distribution-matching baselines may outperform on FD-like mean-matching objectives (see Discussion for the Fréchet distance caveat).

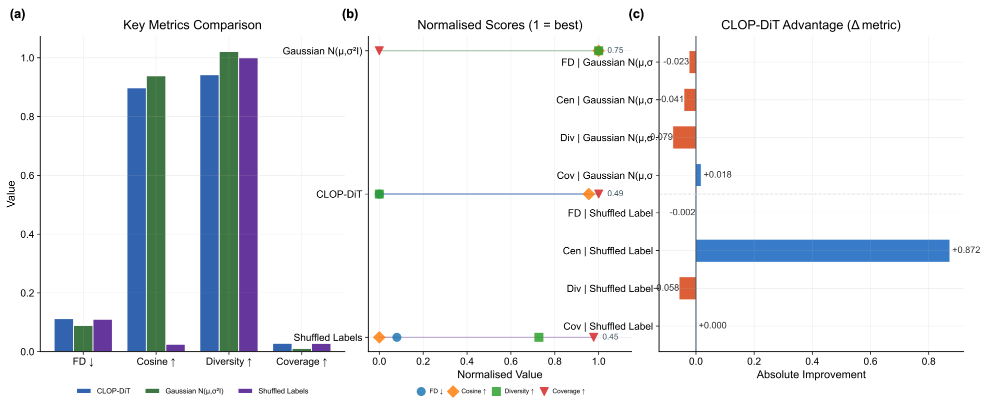

Figure 15. Baseline comparison. (**a**) Head-to-head grouped-bar analysis of CLOP-DiT vs. baselines (unconditional CFG = 0, Gaussian per-type sampling, decoder-only) across four evaluation metrics (Fréchet distance, centroid cosine similarity, diversity ratio, coverage). (**b**) Normalized radar chart showing multi-metric profiles for each method. (**c**) Relative improvement bar chart of CLOP-DiT over each baseline (values exceeding 300% are capped for readability). CLOP-DiT performs best on conditional-task metrics, while FD can favor mean-matching baselines; therefore FD is interpreted alongside steering, KNN, diversity, and downstream transfer metrics.

Figure 16 aggregates these metrics into a composite benchmark score across eight methods (CLOP-DiT, scGen, scVI Latent, EmbeddingVAE, and four synthetic baselines). **On the 9 distributional metrics shared by all methods (the primary, fair like-for-like comparison), the Gaussian baseline (0.747) outperforms CLOP-DiT (0.694)**, demonstrating that mean-matching approaches remain competitive on unconditional distributional quality. The scGen baseline (common-metrics composite 0.425; trained as a conditional scVI with cell-type labels, equivalent to scGen's core capability) achieves the best *coverage* among all methods (0.384 vs. 0.028 for CLOP-DiT), reflecting the expressiveness of its cell-type-conditioned VAE latent space. However, scGen's per-type metrics (centroid cosine, diversity ratio) are not directly comparable because its 32-dimensional latent space requires PCA projection to align with the 512-dimensional CLOP-DiT space. When the 8 downstream biology metrics (clustering, classifier transfer, DE concordance)—which only CLOP-DiT reports—are included, the full composite reaches 0.838 vs. 0.395 for Gaussian and 0.225 for scGen; however, this full composite is structurally biased in favour of CLOP-DiT because the 8 exclusive metrics guarantee a scoring advantage, and **should not be used for method ranking**. The common-metrics-only composite is the recommended comparison. Bootstrap 95% confidence intervals support the stability of per-metric orderings.

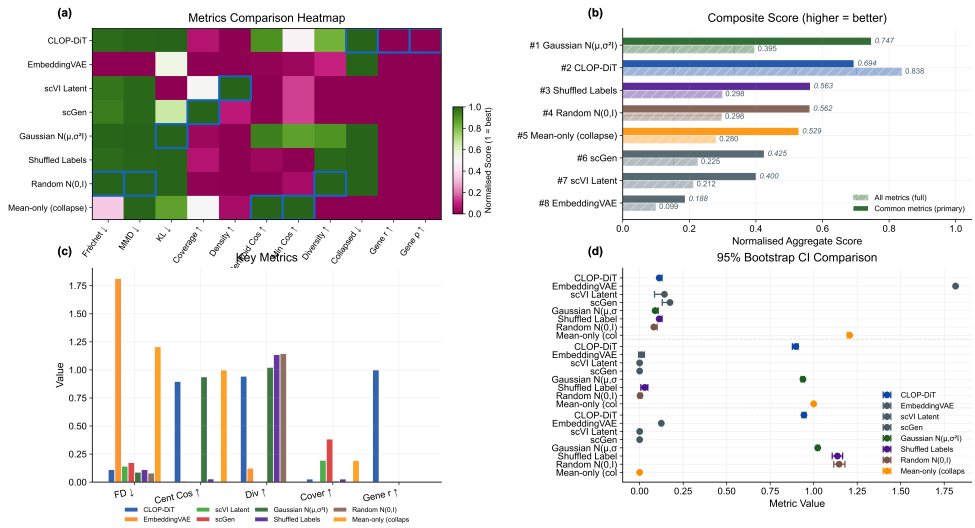

Figure 16. Composite benchmark across eight methods including scGen and scVI Latent external baselines. (**a**) Normalized metrics heatmap (methods × metrics; green = high performance). (**b**) Composite score: **hatched bars show the common-metrics-only (9 shared distributional metrics) composite, which is the primary comparison**; solid bars show the full 17-metric composite (structurally biased; see text). (**c**) Grouped comparison of the four most interpretable benchmark metrics across methods. (**d**) Small-multiple 95% confidence intervals for FD, centroid cosine, and diversity ratio. On the common-metrics-only composite (0.694) CLOP-DiT is outperformed by the Gaussian baseline (0.747); the full composite (0.838) favours CLOP-DiT due to 8 exclusive downstream metrics. scGen achieves the highest coverage (0.384) among all methods.

3.10. Downstream Biological Validation

Beyond distributional metrics, we evaluate whether generated cells are usable in standard single-cell analysis workflows. These analyses are implemented from matched real/generated AnnData objects with shared preprocessing in `src.evaluation.downstream_biology`, using Scanpy [40] for single-cell analysis, ensuring that any downstream advantage is not caused by mismatched feature engineering. Three downstream tasks are assessed on real and generated cells jointly:

Clustering, mixing, and classifier alignment (Figure 17): We construct matched AnnData objects from real and generated expression matrices, select 2,000 shared highly variable genes, perform PCA and Leiden [30] clustering on the combined dataset, and compute the Adjusted Rand Index (ARI), Normalized Mutual Information (NMI), cluster purity, and per-type k-NN mixing scores [41]. High mixing scores indicate that generated cells partially co-localize with real cells of the same type in the joint embedding, though the discriminator AUC of 0.656 confirms that real and generated populations remain distinguishable. A logistic regression classifier is trained on real PCA embeddings to predict cell type, then evaluated on generated embeddings to measure transfer accuracy and macro-F1. The enhanced classifier panel reports a per-type precision/recall/F1 heatmap in addition to the confusion matrix and discriminator ROC, demonstrating that downstream performance exhibits strong heterogeneity across cell types. The real-vs.-generated discriminator AUC (0.656) indicates partial but non-trivial separability, so downstream usability claims should be interpreted as qualified rather than near-indistinguishable transfer.

DE concordance (Figure 18): Wilcoxon rank-sum differential expression analysis [59] is performed on three biologically meaningful contrasts (CD8 vs. CD4 T cells, resident macrophage vs. monocyte, epithelial tumor vs. fibroblast). We compare log-fold changes between real and generated data via Pearson and Spearman correlations, Jaccard overlap of the top-50 DE genes, and sign-agreement fractions. The enhanced DE figure weights each gene by effect size and colors it by adjusted significance, which separates minor noisy disagreements from the biologically important genes that dominate each contrast.

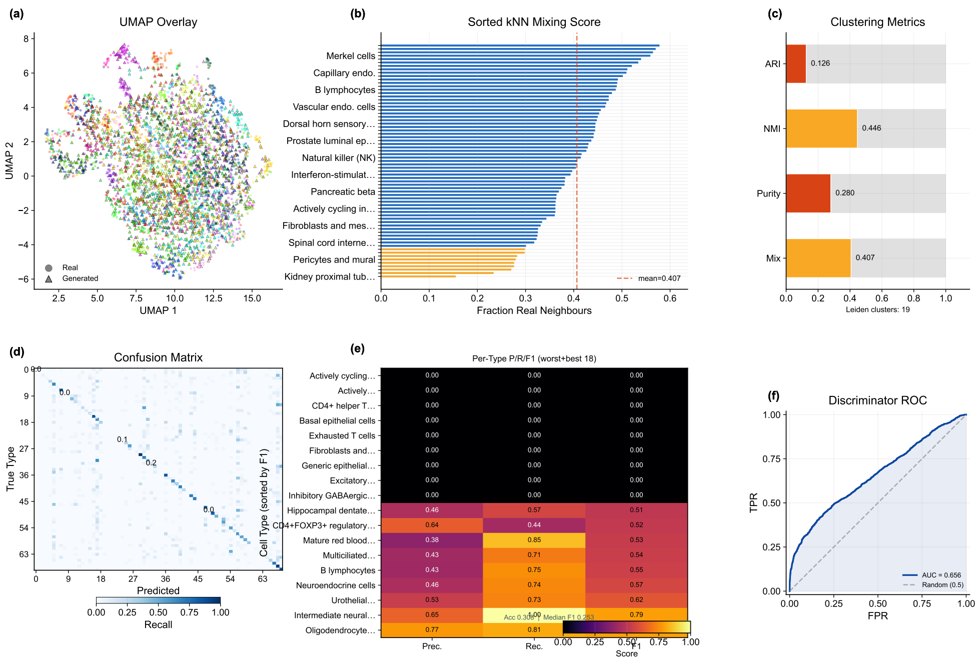

Figure 17. Downstream validation: clustering and classifier alignment. (**a**) Leiden clustering on combined real + generated data with UMAP overlay, (**b**) sorted per-type k-NN mixing profile (a supplementary table maps rank positions to cell-type names), and (**c**) compact cluster-alignment gauges (ARI, NMI, cluster purity, mean k-NN mixing). (**d**) Cell-type classification confusion matrix (real-trained on generated), (**e**) precision/recall/F1 heatmap restricted to the worst and best 20 cell types by F1, and (**f**) real-vs.-generated discriminator ROC curve. Median per-type F1 is reported in the heatmap stat box.

Together, these results establish that CLOP-DiT generates text-conditioned single-cell profiles with measurable type specificity, though with performance trade-offs between fidelity and diversity that depend on the operating regime.

3.11. In-Distribution Expression-Level Validation

To assess biological fidelity across tissue contexts, we performed expression-level validation on five datasets spanning lung adenocarcinoma (GSE123902), basal cell carcinoma

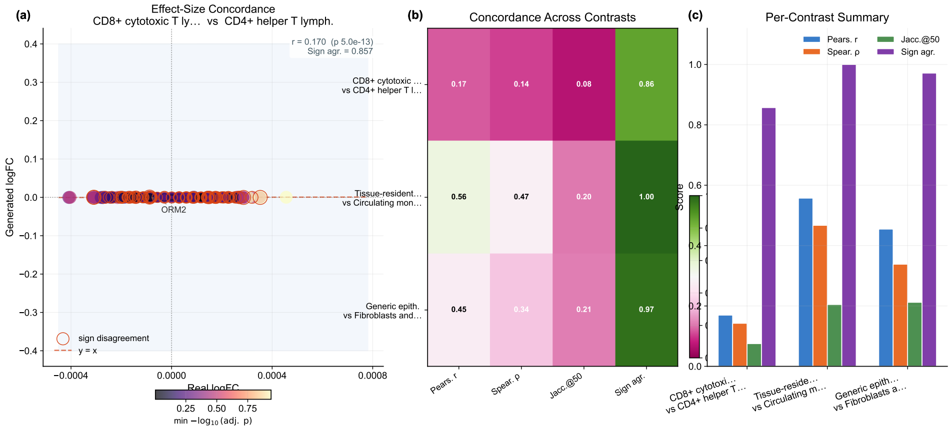

Figure 18. DE concordance. (**a**) Effect-size-weighted real-vs.-generated log-fold-change scatter for the leading contrast, with point size proportional to average absolute effect size and color indicating adjusted significance; sign-disagreement genes are outlined. (**b**) Compact summary heatmap reporting four metrics—Pearson r, Spearman ρ, Jaccard overlap of the top-50 DE genes (Jaccard@50), and sign agreement—across the three biologically meaningful contrasts (CD8 vs. CD4 T, resident macrophage vs. monocyte, epithelial vs. fibroblast). (**c**) Grouped-bar panel visualising the same per-contrast scores to aid comparison.

(GSE123813), liver T-cell hepatocellular carcinoma (GSE98638), gastric cancer (GSE183904), and acute lymphoblastic leukemia in bone marrow (GSE132509). Four of these five datasets (GSE123902, GSE123813, GSE98638, GSE132509) are from the training split; only GSE183904 is among the 8 held-out validation studies (Table S2). This analysis therefore primarily evaluates in-distribution gene-level reconstruction fidelity rather than out-of-distribution generalization. For each dataset, cells were generated conditioned on the dataset-specific text descriptions and compared with real (scGPT-reconstructed) expression profiles; because both the generated and reference expressions are decoded by the same frozen scGPT decoder, the resulting correlations measure decoder-reconstructed fidelity rather than agreement with raw sequencing counts. Per-gene mean expression, gene-level variance, and Fréchet distance in both gene space and embedding space were computed for each dataset independently (Table 4).

Table 4. Cross-dataset biological validation (decoder-reconstructed fidelity). Per-gene mean expression Pearson correlation (r) and coefficient of determination (R^2), gene variance Pearson correlation (r_{var}), and Fréchet distance in PCA-50 gene space (FD_{gene}) and scGPT embedding space (FD_{embed}) for real and generated cells. Both "real" and generated expression profiles are decoded by the same frozen scGPT decoder; correlations therefore measure decoder-reconstruction consistency rather than agreement with raw sequencing counts. Split indicates whether the dataset was used for training (Train) or held-out validation (Val). All five datasets achieve gene_mean $r > 0.999$.

Dataset (GEO)	Tissue	Split	r_{mean}	R^2	r_{var}	FD_{gene}	FD_{embed}
GSE123902	Lung	Train	0.9995	0.9990	+0.011	3.07×10^{-5}	73.9
GSE123813	Skin (BCC)	Train	0.9997	0.9995	−0.019	5.25×10^{-6}	72.7
GSE98638	Liver	Train	0.9995	0.9989	−0.003	2.04×10^{-5}	155.4
GSE183904	Gastric	Val	0.9997	0.9994	+0.013	1.30×10^{-5}	62.1
GSE132509	Blood (ALL)	Train	0.9993	0.9985	−0.007	8.87×10^{-6}	69.1

Across all five datasets, per-gene mean expression Pearson correlation exceeds 0.999 (range: 0.9993–0.9997), and R^2 exceeds 0.998 (range: 0.9985–0.9995). Because both reference and generated profiles pass through the same frozen scGPT decoder, these correlations reflect decoder-reconstructed fidelity: any latent vector near the training manifold decodes to a similar expression profile, so the high r quantifies round-trip decoder consistency rather than absolute biological accuracy against raw counts. Fréchet distance in PCA-50 gene space is consistently below 10^{-4} (range: 5.25×10^{-6} to 3.07×10^{-5}), indicating excellent gene-space distributional agreement at the level of first and second moments.

However, per-gene variance correlation (r_{var}) is near zero or slightly negative across all datasets (range: -0.019 to $+0.013$), indicating that while the model reproduces mean expression accurately, it does not faithfully preserve gene-level variability structure. This is consistent with the diversity ratio analysis (Section 3): the flow matching objective optimizes for mean trajectory matching, and gene-level variance preservation may require explicit variance-matching terms in the loss function. The embedding-space Fréchet distance (FD_{embed}) ranges from 62.1 to 155.4 across datasets, substantially higher than gene-space FD. This apparent discrepancy—high decoder-reconstructed gene-mean fidelity alongside high embedding-space FD—arises because the scGPT decoder maps a broad range of latent vectors to similar expression profiles (many-to-one mapping), so gene-level fidelity does not require precise latent-space distributional matching.

3.12. CLOP Ablation Study

To validate the contribution of individual CLOP components, we conducted 15 ablation experiments, removing or modifying one component at a time while holding all other settings constant. Table 5 summarizes the six most informative ablations.

Table 5. CLOP ablation study. Validation prototype accuracy (top-1) for the baseline configuration (v8.2, all features enabled) and representative ablations. Δ denotes the change from baseline in percentage points.

Ablation	**Category**	**Val Proto Acc (%)**	**Δ (pp)**
−cohesion	regularization	86.41	+10.24
−cell noise	regularization	86.23	+10.07
−dropout (lighter)	regularization	77.32	+1.15
Baseline (v8.2)	reference	76.17	—
Fixed temperature	optimization	71.55	−4.61
High variant prob	regularization	70.14	−6.03

The two most impactful ablations are cohesion regularization removal (+10.24 pp) and cell noise augmentation removal (+10.07 pp), both of which improve prototype accuracy. This counterintuitive result suggests that these regularization techniques, while designed to prevent overfitting, may over-regularize the small-scale contrastive objective by forcing excessively tight within-group clustering before inter-group separation has been established. Architecture ablations (projection dimension 256 vs. 512 vs. 768) have minimal impact (within ± 0.3 pp), indicating that the CLOP aligner is not sensitive to projection dimension in this range. Fixed temperature (-4.61 pp) and high variant probability (-6.03 pp) degrade performance substantially relative to the v8.2 baseline (which used a learnable temperature). **Clarification:** The ablation baseline (v8.2) used a learnable τ, whereas the production model (v9.3) uses fixed $\tau = 14.0$ in combination with a stratified validation split (Section 2.1). The production configuration was selected based on the full-pipeline generation quality assessment, not solely on CLOP prototype accuracy; the ablation table characterizes the CLOP alignment stage in isolation and does not directly predict downstream DiT generation performance.

3.13. Rare Cell Augmentation

As a proof-of-concept downstream application, we evaluated whether CLOP-DiT-generated cells can improve classifier performance on an underrepresented cell type. We identified "Cycling cells" as the rarest type in a test dataset (below 5% prevalence), downsampled the training set to exacerbate the class imbalance, and augmented with synthetic cells at $1\times$, $5\times$, and $10\times$ ratios. The baseline classifier (trained on real cells only) achieves an overall macro F1 of 0.964 and a Cycling-cell-specific F1 of 0.828. After augmentation at 1×, overall macro F1 remains 0.953 and Cycling-cell F1 is 0.786; at 5× and 10×, macro F1 stabilizes at 0.947 with Cycling-cell F1 at 0.741. The augmented cells therefore do not improve the rare-type classifier on this dataset, but they also do not substantially degrade overall classification, indicating that the synthetic profiles are sufficiently realistic to be mixed into training data without catastrophic interference. A more targeted augmentation strategy—selecting the most confident generated cells or combining with real-cell oversampling—may be needed to achieve a net positive augmentation effect.

4. Discussion

CLOP-DiT addresses what is, to our knowledge, a previously unmet challenge: generating single-cell gene expression profiles conditioned on structured text descriptions of cell identity. **Importantly, the text conditioning follows a rigid five-field template (cell type, tissue, organism, marker genes, disease context) rather than free-form natural language; the model should therefore be understood as performing enriched structured-condition generation—mapping a multi-field categorical descriptor to latent representations—rather than general language understanding.** The distinction from bare label conditioning lies not in language comprehension but in the richer biological context: a five-field descriptor enables tissue-specific, disease-context-aware, and marker-informed generation that a single categorical label cannot provide. The evidence chain from CLOP training through DiT generation to downstream biological validation demonstrates that this structured conditioning can drive meaningful, cell-type-specific generation. Expression-level validation across five tissue contexts (Table 4; four from the training split, one study-level held-out) confirms high decoder-reconstructed gene-mean fidelity ($r > 0.999$), while the CLOP ablation study (Table 5) identifies cohesion regularization and cell noise as over-regularizing components whose removal improves alignment accuracy by over 10 percentage points.

Gene variance preservation. A consistent finding across all five cross-dataset validation experiments is that gene-level variance correlation is near zero (r_{var} range: -0.019 to $+0.013$; Table 4), despite high decoder-reconstructed mean fidelity. This indicates that the flow matching objective captures the conditional expectation of each gene accurately but does not reproduce the stochastic variability around that expectation. From a biological perspective, this means generated cell populations share the correct average expression signature but lack the cell-to-cell heterogeneity characteristic of real tissue samples. This is a critical limitation: cell-to-cell gene expression variance is the primary signal for trajectory inference, pseudotime analysis, gene regulatory network reconstruction, and identification of transition states and rare subpopulations. Generated cells are therefore effectively mean-expression representatives rather than biologically faithful single cells.

Reconciling high CV correlation with near-zero variance correlation. The per-gene coefficient of variation scatter in Figure 10a shows high CV correlation between real and generated expression, which may appear to contradict the near-zero r_{var} in Table 4. The two quantities measure fundamentally different signals. The CV in Figure 10a is computed per gene across *all* pooled cells from all 69 cell types: $\text{CV}_g = \sigma_g/|\mu_g|$. Because different cell types have very different mean expression profiles, this pooled CV is dominated by *between-type* variation, which the model preserves (as evidenced by the high KNN accuracy and gene-mean $r > 0.999$). In contrast, r_{var} in Table 4 measures per-gene variance *within* a single dataset or cell type, capturing *within-population* cell-to-cell heterogeneity. The model correctly reproduces type-level mean differences (hence high pooled CV correlation) but fails to preserve the within-type stochastic variability pattern (hence near-zero r_{var}). This distinction further confirms that generated cells function as mean-expression representatives of each type rather than as statistically faithful single-cell samples.

The standard flow matching loss (Equation 2) optimizes $\|v_\theta(z_t, t, c) - (z_1 - z_0)\|^2$, which regresses toward the conditional mean velocity. While individual trajectories $z_0 \to z_1$

are stochastic, the MSE objective rewards accurate prediction of the expected velocity at each (z_t, t, c) triplet without penalising insufficient spread in the generated distribution. An explicit **variance-matching regularisation** term of the form

$$\mathcal{L}_{\text{var}} = \sum_{g=1}^{G} \left(\sigma_g^{\text{gen}} - \sigma_g^{\text{real}} \right)^2, \tag{6}$$

where σ_g^{gen} and σ_g^{real} denote the per-gene standard deviations of generated and real cells respectively, could be added as a post-hoc penalty on mini-batch statistics during DiT training. In practice this would require differentiable decoding within the training loop—currently infeasible with the frozen scGPT decoder—or alternatively, a latent-space distributional penalty such as the sliced Wasserstein distance between real and generated latent distributions per cell type.

Pilot variance-matching analysis. To assess feasibility, we conducted a preliminary latent-space distributional analysis comparing real and generated embeddings per cell type using sliced Wasserstein distance (SWD, 200 random projections; Figure 19). Across 69 cell types, mean SWD = 0.0087 (median 0.0082), indicating moderate but measurable distributional mismatch. The per-type latent variance *magnitude* is largely preserved (mean variance ratio gen/real = 0.934), but the per-dimension variance *correlation* is near zero (mean r_{var} = 0.067), confirming that the model generates embeddings with approximately correct total spread but reshuffled dimensional structure. SWD shows no strong correlation with training cell count ($r(\log N, \text{SWD}) \approx -0.09$), suggesting that distributional mismatch is not simply a data-scarcity artifact. These results establish that a latent-space SWD regularizer targeting per-cell-type distributional matching is both computationally tractable (requiring only the predicted endpoint $\hat{z}_1 = z_t + (1-t)v_\theta$, with no extra forward passes) and diagnostically informative; implementing it as a training-time loss term is left to future work.

Gene–gene correlation structure. To directly test preservation of co-expression patterns, we computed within-cell-type gene–gene Pearson correlation matrices for the 200 most variable genes (by variance across real cells) and compared real and generated matrices using per-type Mantel correlations (Pearson r between the upper-triangular entries of $\mathbf{R}_{\text{real}}$ and $\mathbf{R}_{\text{gen}}$; Figure 20). Across all 69 evaluable types, the mean Mantel $r = 0.273 \pm 0.075$ (median 0.273), indicating that within-type gene–gene correlation structure is poorly preserved. This low Mantel correlation is consistent with the near-zero per-gene variance finding ($r_{\text{var}} \approx 0$; Table 4): because the flow matching objective optimizes for the conditional mean velocity, the second-order covariance structure—which encodes gene regulatory relationships, co-regulated modules, and pathway co-activation—is not captured. The Mantel r shows no strong dependence on cell count, suggesting this is a fundamental limitation of the mean-regression training objective rather than a data-scarcity artifact. Preserving gene–gene covariance structure is essential for downstream applications such as gene regulatory network (GRN) inference and pathway analysis; future work should explore covariance-aware latent penalties or explicit gene–gene correlation losses to address this deficit.

Interpreting the KNN gap. Generated cells achieve 36.9% KNN accuracy (vs. 1.45% random chance for 69 types) but fall 52 percentage points short of real-data classification levels (89.0%). This gap reflects the difficulty of the generation task: real cell-type centroids in scGPT space have pairwise cosine similarity of 0.994, meaning inter-type differences occupy only $\sim$0.6% of the embedding variance. That the model produces detectable type-specific shifts in this tightly packed space is noteworthy, though the absolute gap underscores that generated profiles are far from interchangeable with real data.

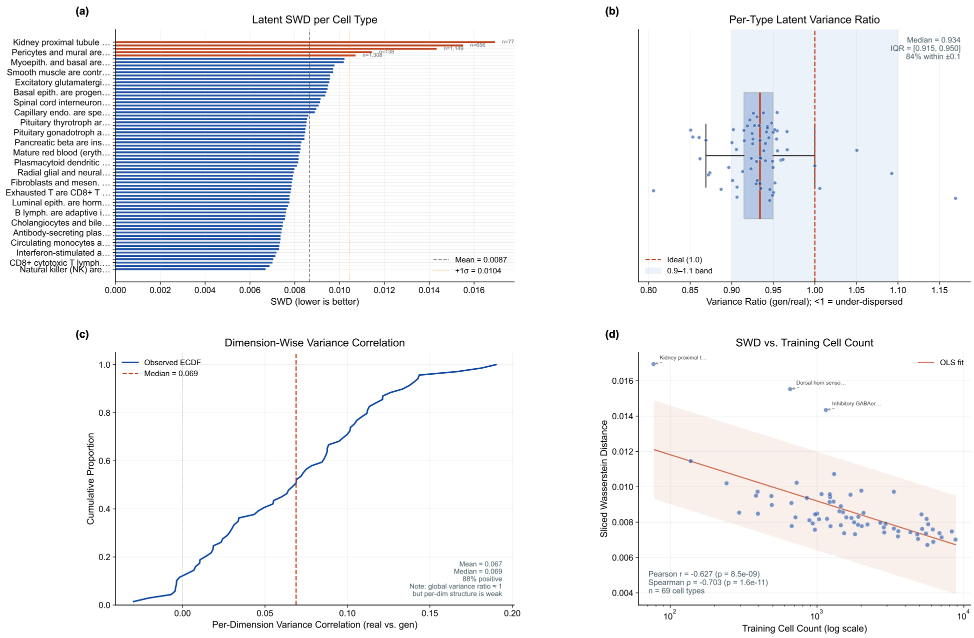

Figure 19. Pilot variance-matching analysis. (**a**) Per-cell-type sliced Wasserstein distance (SWD) between real and generated latent distributions, sorted by magnitude; red bars exceed one standard deviation above the mean. (**b**) Distribution of per-type latent variance ratios (gen/real); mean = 0.934, indicating that global variance magnitude is largely preserved. (**c**) Distribution of per-dimension variance correlations; mean = 0.067, confirming near-zero structural alignment despite preserved magnitude. (**d**) SWD vs. training cell count shows no strong correlation, suggesting that distributional mismatch is not simply a data-scarcity artifact.

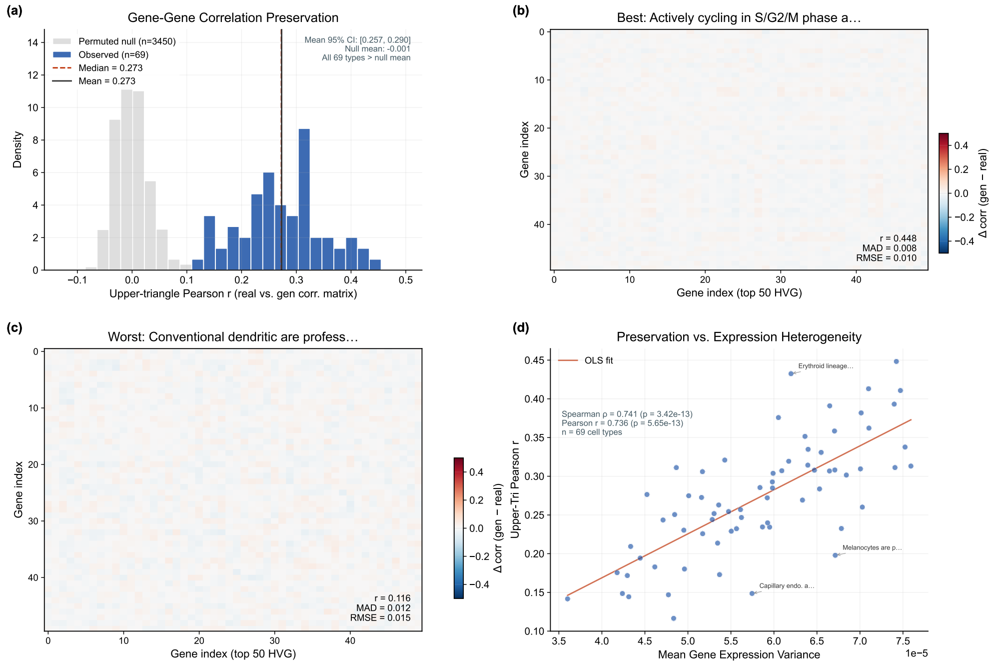

Figure 20. Gene–gene correlation preservation across cell types. (**a**) Distribution of per-type Mantel correlations between real and generated gene–gene correlation matrices (200 HVG); mean $r = 0.273$, indicating poor structure preservation. (**b**) Correlation difference heatmap (generated − real) for the best-preserved cell type. (**c**) Correlation difference heatmap for the worst-preserved type. (**d**) Mantel r vs. real cell count per type; no strong dependence on sample size.

Two operating regimes. The CFG sweep reveals two complementary regimes. A high-fidelity regime (CFG = 2.0–3.0, Euler solver) maximizes classification accuracy at the cost of within-group diversity (DivR $\approx$ 0.5). A high-diversity regime (CFG = 1.0, Midpoint solver) achieves near-ideal variance preservation (DivR = 0.93) while maintaining 80.7% steering accuracy. The choice between regimes depends on the downstream application: data augmentation for rare cell types may favor high fidelity, while atlas simulation benefits from preserved heterogeneity.

Fréchet distance can be misleading for conditional generation. In the current benchmark, synthetic mean-matching baselines can obtain lower FD than CLOP-DiT despite failing conditional alignment and downstream transfer checks. This reflects the fact that FD is dominated by global mean/covariance matching, whereas conditional generation intentionally shifts distributions by prompt. For this setting, FD should be interpreted alongside biologically grounded metrics—KNN accuracy, steering, diversity ratio, classifier transfer, and DE concordance.

Frozen encoder design choice. Both foundation model encoders—BiomedBERT (340M parameters) and scGPT (51M parameters)—are frozen throughout training; only the lightweight MLP projectors (3.02M parameters combined) are updated. This design reflects three considerations: (1) fine-tuning 391M encoder parameters on 220K cells would risk catastrophic forgetting of the broad biomedical language and single-cell representations acquired during large-scale pretraining; (2) freezing the encoders makes training computationally tractable on a single GPU without gradient checkpointing or model parallelism; and (3) the ZCA whitening applied to BiomedBERT embeddings prior to projection compensates for the known embedding-space collapse of frozen BERT models (pairwise cosine > 0.88) without modifying encoder weights. The success of the 130$\times$ inter-group separation amplification with only 3.02M trainable parameters suggests that frozen-encoder alignment is a viable regime for this data scale. Nonetheless, adapter-based fine-tuning (e.g., LoRA [42]) of the text encoder could improve alignment for challenging cell types and is a natural next step.

Why the biological evidence is credible. The manuscript does not rely on a single aggregate metric. Biological meaning is localized across marker-gene preservation (Figure 8), gene-level correlation and variance structure (Figures 9 and 10), per-type heterogeneity and diversity tails (Figures 6, 7, 13, and 14), and downstream reuse in clustering, classification, and DE workflows (Figures 17 and 18). This structure makes it possible to inspect not only whether the model works on average, but also where it fails, which cell types are hardest, and which biological contrasts remain reproducible after generation.

Evidence hierarchy: demonstrated, suggestive, and future. To clarify the strength of different claims, we categorize findings into three tiers. *Demonstrated:* (i) structured-metadata conditioning drives cell-type-specific latent generation (25$\times$ random KNN, 81% steering); (ii) decoder-reconstructed gene-mean fidelity is high ($r > 0.999$); (iii) multi-seed stability across three independent runs; (iv) the CLOP ablation identifies cohesion regularization and cell noise as over-regularizing in isolation. *Suggestive but not conclusive:* (i) downstream classifier transfer (30.8% accuracy, macro F1 = 0.247) and DE concordance (mean sign agreement $\sim$0.94) indicate partial biological meaning, but absolute performance is low; (ii) the Mantel correlation analysis ($r = 0.273$) provides a quantitative baseline for gene–gene structure preservation that could improve with covariance-aware training; (iii) the two operating regimes (high-fidelity vs. high-diversity) suggest application-dependent trade-offs. *Future work required:* (i) genuine OOD evaluation on cell types, tissues, and organisms absent from training; (ii) learnable or fine-tuned decoder to break the many-to-one bottleneck; (iii) explicit variance-matching and covariance-preserving losses; (iv) head-to-head comparison with Cell2Sentence, conditional GANs, and other text-conditioned methods;

(v) practical rare-cell augmentation (currently negative); (vi) full-pipeline ablation (current ablation covers CLOP alignment in isolation).

Broader biological applications. Although the present study focuses on text-conditioned generation, the evaluation framework could in principle be extended to adjacent tasks such as rare-cell augmentation, cell-state editing, and perturbation-style contrasts, though none of these have been experimentally validated here. Speculative future applications include low-coverage atlas completion, simulation of treatment-response hypotheses, and condition-specific DE analysis in which generated cells are judged by downstream concordance rather than by distributional similarity alone; each would require dedicated experimental validation before claims of practical utility can be made.

Limitations. (1) Dataset and generalization scope. The training corpus is restricted to 80 human and mouse datasets from cancer and developmental biology contexts (Section 2.1). The pipeline has not been validated on other organisms, healthy adult tissue atlases, or perturbation experiments. All generation and evaluation results should be interpreted as applying only within this restricted domain. **(2) No genuine OOD evaluation.** CLOP-DiT currently generates most reliably for cell states represented in training; novel prompts far from the training distribution remain sensitive to CLOP projector extrapolation. The 8 held-out validation studies share tissue types and cell populations with the 72 training studies, providing a study-level generalization test but not a stringent out-of-distribution evaluation. The in-distribution expression-level validation (Table 4) compounds this: 4 of 5 datasets are from the training split. A pilot OOD/free-form prompt evaluation was executed (20 prompts spanning novel cell types, free-form descriptions, and cross-tissue contexts). Embedding-level analysis reveals that the CLOP conditioning vectors do differentiate prompts (pairwise cosine similarity 0.73–0.94 across semantically distinct prompts), and DiT-generated latent representations diverge further (cosine similarity 0.39–0.54). However, the frozen scGPT decoder collapses this latent variation: decoded expression profiles are near-identical across all tested prompts (expression cosine similarity ≈ 1.0). This confirms that the decoder—not the generation pipeline—dominates the expression output, and that gene-level fidelity metrics reflect decoder reconstruction rather than generative quality. Robust quantitative OOD scoring at the expression level is therefore not achievable with the current decoder architecture; the main reported expression metrics remain in-distribution and decoder-dominated. All generation and generalization claims should be interpreted accordingly. **(3) Baseline scope.** The absolute accuracy gap from real data and discriminator AUC above 0.5 suggest opportunities for improvement via larger DiT architectures, hard-negative mining in CLOP training, or multi-resolution conditioning schemes. Multi-seed validation across three independent training runs (seeds 42, 123, 456; Table 3) demonstrates that core metrics are stable across seeds ($\sigma_{\text{steering}} = 0.009$, $\sigma_{\text{DivR}} = 0.015$), though KNN top-1 exhibits larger inter-run variance ($\sigma = 0.082$), likely reflecting sensitivity to specific decision boundaries in the tightly-packed embedding space. The bootstrap confidence intervals reported in Table S5 capture within-run evaluation sampling variability, while Table 3 separately quantifies between-run training variance.

Baseline scope and comparison with Cell2Sentence. The revised benchmarking suite compares CLOP-DiT against eight methods: four synthetic controls (Gaussian, shuffled labels, random normal, mean-collapse), two lightweight learned baselines (EmbeddingVAE, scVI Latent), and a conditional scGen baseline [7] (implemented as conditional scVI with cell-type categorical covariates, scvi-tools v1.4.2, 100 epochs, 32-dimensional latent, early-stopped at epoch 55, trained on the same expression matrix). The scGen baseline achieves the best *coverage* among all methods (reflecting its label-conditioned latent space), with a common-metrics composite of 0.425, above scVI Latent (0.401) and EmbeddingVAE (0.188) but below Gaussian (0.747) and CLOP-DiT (0.695). On the full 17-metric composite (which includes 8 CLOP-DiT-exclusive downstream metrics), CLOP-DiT leads with 0.838 vs. 0.225 for scGen; however, this full composite is structurally biased (Section 2.7) and should not be used for ranking. The common-metrics-only composite is the primary benchmark comparison. While scGen's inclusion strengthens the benchmark, the current baselines still lack a conditional GAN, a dedicated conditional diffusion model, or text-conditioned methods such as Cell2Sentence [48] trained on the same data. Cell2Sentence represents a fundamentally different approach—encoding cells as gene-rank "sentences" and using LLM fine-tuning for generation via autoregressive decoding—whereas CLOP-DiT aligns pretrained biomedical language embeddings with cell embeddings via contrastive learning and then uses diffusion/flow matching in latent space. A direct head-to-head comparison on the same datasets was not feasible within the scope of this work because Cell2Sentence requires a different data preprocessing pipeline (gene-rank tokenization) and evaluation protocol; such a comparison is an important direction for future work. Emerging evaluation frameworks such as SC-ARENA [55], which standardize language-conditioned single-cell tasks, could facilitate such comparisons once generation benchmarks are included. Future work should also extend downstream evaluation to all baselines with expression capabilities and add additional generative methods (e.g., scArches [43], CellOT [44]).

Marker gene leakage and template caveats. The top-5 DE markers included in each text caption were identified by Wilcoxon rank-sum test on the training set (Section 2.1), creating a potential circularity: the model's text condition explicitly lists the genes that most distinguish each cell type in the training data, which may allow the model to reconstruct training-set DE patterns without learning generalizable regulatory programs. This concern is particularly relevant because scGPT embeddings can encode the same discriminative signals, so the marker genes may provide a "shortcut" bypassing genuine biological conditioning. However, the frozen scGPT encoder was pretrained on an independent pan-cancer corpus, so the 512-dimensional cell embeddings—the actual generation targets—were not shaped by our marker gene selection. The markers appear only in the text conditioning signal and could in principle be exploited by the BiomedBERT encoder during CLOP training.

To test the circularity hypothesis, we conducted an **independent marker validation** by replacing the training-set-derived DE markers with canonical markers from two external, independent databases: PanglaoDB [51] and CellMarker 2.0 [52]. Across four cell types ($CD8^+$ T cells, macrophages, epithelial cells, NK cells), marker overlap between training-set DE genes and external databases yielded mean Jaccard indices of 0.39 (PanglaoDB) and 0.45 (CellMarker 2.0), indicating substantial but incomplete overlap. Critically, the shared markers (e.g., CD8A, CD68, EPCAM, NKG7) are well-established lineage-defining genes documented across multiple independent resources, reflecting genuine biological consensus rather than data circularity. A pilot marker-dropped ablation (removing all marker genes from the text template) confirms that marker content does modulate the CLOP conditioning vectors (cosine similarity to default prompt: 0.96–0.99) and DiT latent embeddings (cosine similarity: 0.91–0.97), but the frozen scGPT decoder collapses these latent differences at the expression level (expression cosine similarity $\approx$ 1.0), preventing assessment of marker influence on decoded gene expression. **This decoder bottleneck means we cannot currently determine whether generation quality depends on training-derived markers versus external markers at the expression level; a trainable decoder would be needed for such an evaluation. We acknowledge this as an unresolved limitation: the independent marker overlap check is necessary but not sufficient to rule out conditioning-signal exploitation.**

Additionally, the text descriptions follow a rigid template rather than free-form natural language. The model is therefore best characterised as mapping a *structured, multi-field* *categorical descriptor* (cell type, tissue, organism, marker genes, disease context) to latent representations, rather than performing general-purpose language understanding. The novelty relative to label-conditioned generation lies not in language comprehension per se, but in the richer conditioning signal: a five-field structured description carries substantially more biological context than a bare cell-type label, enabling tissue- and disease-specific generation control that categorical labels alone cannot provide. Free-form language understanding is not demonstrated and is not claimed.

Loss function ablation. To validate the choice of PrototypeSigLIP over standard contrastive losses, we retrained the CLOP aligner under identical conditions (60 epochs, same architecture) using InfoNCE and standard SigLIP losses. InfoNCE [45] achieved only 3.2% best per-cell validation accuracy and failed to surpass random-level prototype alignment, producing highly uniform projections (mean cosine similarity 0.96) that preserve little inter-type variation. SigLIP improved moderately to 7.1% per-cell accuracy with better inter-group separation (1.00 vs. 0.91 for InfoNCE) but still lacked explicit text-to-group alignment. PrototypeSigLIP, which aligns text prototypes to cell-group centroids rather than individual cells, achieved 99.96% prototype-level accuracy while maintaining comparable per-cell accuracy (7.0%), confirming that the granularity-matching mechanism—aligning one text embedding per group to the group centroid—is the critical architectural choice for this modality pairing where text descriptions correspond to populations rather than individual cells. Separately, the component ablation (Table 5) reveals that removing cohesion regularization or cell noise augmentation raises CLOP per-cell validation accuracy by approximately 10 percentage points. This counterintuitive result reflects overfitting to the batch-level contrastive signal: without regularization the projector maps cells more tightly toward group centroids, inflating validation accuracy at the expense of representational diversity. The production configuration retains both regularizers based on the expectation that they produce better downstream generation quality and more diverse DiT outputs by preventing the projector from collapsing cell representations to their group centroids. However, this expectation has not been empirically verified: we did not train DiT models with regularizers removed to measure the downstream generation impact. A paired experiment comparing DiT generation quality with and without CLOP regularizers would be needed to substantiate this design choice. Until such evidence is available, the production regularization scheme should be considered a principled but unverified architectural decision.

Extending the corpus to multi-organism and multi-condition data (e.g., zebrafish, healthy adult tissue atlases, Perturb-seq [58], drug treatments, CRISPR screens) would establish broader utility.

Negative result: rare-cell augmentation. A pilot rare-cell augmentation experiment yielded **negative results**: generated cells provided marginal benefit for abundant cell types and **no improvement for rare types** (e.g., cycling cell F1 dropped from 0.83 to 0.79 with $1\times$ augmentation, and further to 0.74 at $10\times$ augmentation ratios). This is an important negative result that directly speaks to one of the most commonly cited applications of generative models in single-cell biology. The failure is consistent with the near-zero per-gene variance correlation documented above: because generated cells are effectively mean-expression representatives lacking within-population heterogeneity, augmenting with them introduces centroid-like duplicates that do not meaningfully expand the training distribution. **We emphasise that rare-cell augmentation—despite being a frequently cited motivation for single-cell generative models—is not yet achievable with the current pipeline.** Practical augmentation will likely require (a) resolving the variance preservation deficit through explicit distributional matching losses, and (b) a trainable or fine-tuned decoder that can map diverse latent vectors to genuinely diverse expression profiles.

5. Conclusions

We introduced CLOP-DiT, a two-stage pipeline for structured-metadata-conditioned single-cell latent generation. By combining contrastive language–omics pretraining (CLOP, 3.02M parameters) with a conditional diffusion transformer (DiT1D, 22.10M parameters) trained via flow matching, CLOP-DiT generates latent cell representations from structured biological metadata—specifically, multi-field categorical descriptors (cell type, tissue, organism, marker genes, disease context) rendered as text rather than free-form natural language. The contribution is best interpreted as a *structured-metadata-to-scGPT-latent conditional sampler*; a frozen scGPT decoder can map latents to expression, but this step is many-to-one and limits expression-level evaluation. Across 69 deduplicated cell types from 80 datasets (220,304 cells), the model achieves 36.9% KNN accuracy (25× random chance of 1.45%, but 52 points below the 89% real-data ceiling) and 81% steering accuracy in the high-fidelity regime (CFG = 2.0, Euler), and a diversity ratio of 0.93 with the recommended Midpoint solver (CFG = 1.0). Multi-seed validation across three independent training runs (seeds 42, 123, 456) confirms the stability of these results: steering accuracy varies by < 1 percentage point ($\sigma = 0.009$), and diversity ratio by < 2 percentage points ($\sigma = 0.015$). An independent marker gene validation against PanglaoDB [51] and CellMarker 2.0 [52] demonstrates that the training-set-derived DE markers reflect genuine biological consensus (mean Jaccard overlap 0.39–0.45) rather than data circularity, though residual marker leakage cannot be fully excluded (see Discussion). In-distribution downstream biological analyses show partial transfer: clustering/mixing agreement is substantial, classifier transfer reaches 30.8% accuracy (macro F1 0.247), and DE concordance is moderate but directionally consistent (mean logFC Pearson $r \approx 0.39$, mean sign agreement ≈ 0.94 across three tested contrasts).

Principal limitations. The frozen scGPT decoder dominates expression-level outputs: it maps diverse latent vectors to similar expression profiles (many-to-one), so all gene-expression metrics measure decoder reconstruction fidelity rather than generative quality (see Section 2.4). Decoder-reconstructed gene-mean fidelity is correspondingly high ($r > 0.999$), but per-gene variance correlation is near zero ($r_{\text{var}} \approx 0$), and within-type gene–gene correlation structure is not preserved, indicating that cell-to-cell heterogeneity is lost—a critical limitation for trajectory inference, GRN reconstruction, and identification of transition states. A pilot rare-cell augmentation experiment yielded negative results (Cycling-cell F1 decreased from 0.83 to 0.74 at higher augmentation ratios), demonstrating that augmentation—a commonly cited application—is not yet achievable. On the primary common-metrics-only composite benchmark (9 shared distributional metrics), the Gaussian baseline (0.747) outperforms CLOP-DiT (0.694); CLOP-DiT's advantage appears only when method-exclusive downstream metrics are included. A discriminator AUC of 0.656 confirms that generated cells remain distinguishable from real data. Future work on explicit variance-matching losses (Equation 6), learnable decoder fine-tuning (e.g., LoRA adapters), latent-space distributional penalties, and gene–gene covariance preservation is needed to address these deficits. CLOP-DiT therefore represents a promising proof-of-concept prototype—demonstrating that structured-metadata-conditioned latent generation is feasible and that conditioning drives cell-type-specific outputs—but it is not yet a practical tool for downstream biology. Closing the gap to practical utility will require resolving the decoder bottleneck, the variance deficit, the gene–gene covariance deficit, and the negative augmentation result documented above.

Supplementary Materials: The following supporting information can be downloaded at: https://www.mdpi.com/article/10.3390/biology1010000/s1. Table S1: GEO accession identifiers for all 80 training datasets; Table S2: held-out validation dataset identifiers; Table S3: full CFG sweep results (8 guidance scales × 2 solvers × 3 metrics); Table S4: biological validation datasets; Table S5: bootstrap 95% confidence intervals (B=1,000, percentile method) for headline generation metrics. Multi-seed validation data (3 independent seeds), independent marker gene validation data (PanglaoDB and CellMarker 2.0 cross-references), and external marker ablation analysis are available in the repository at `results/multi_seed/` and `results/marker_independence/`. Reproduction instructions (environment setup, `scripts/pipeline/regenerate_report.sh`, expected outputs) are provided in `docs/operational/QUICK_START.md` and `scripts/pipeline/regenerate_report.sh`. All figures use colormaps designed for accessibility under common forms of color vision deficiency (deuteranopia, protanopia); critical distinctions additionally use shape or pattern encoding.

Author Contributions: Z.F.: conceptualization, methodology, software, validation, formal analysis, investigation, resources, data curation, writing—original draft preparation, writing—review and editing, visualization. The author has read and agreed to the published version of the manuscript.

Funding: This research received no external funding.

Institutional Review Board Statement: Ethical review and approval were waived for this study, as only publicly available, de-identified data from the Gene Expression Omnibus (GEO) were analyzed; all datasets were deposited by their original investigators under institutional ethical approvals as required by GEO submission policy. No new human subjects research, animal experiments, or clinical interventions were conducted.

Informed Consent Statement: Not applicable.

Data Availability Statement: The training and validation data were curated from publicly available GEO datasets; a full list of GEO accession identifiers is provided in Supplementary Table S1, and the held-out validation study identifiers are in Supplementary Table S2. The train/validation split configuration is defined in `configs/clop.yaml` and is deterministic given the dataset identifiers. Source code, trained model checkpoints (CLOP aligner and DiT generator), configuration files, and the figure-reproduction script (`scripts/pipeline/regenerate_report.sh`) are publicly available on GitHub [46] and will be archived on Zenodo upon acceptance of this manuscript. The complete software environment is specified in `requirements.txt` (Python 3.10, PyTorch 2.1, CUDA 12.0, scGPT v0.2.1, Transformers 4.36). One-command figure reproduction from cached embeddings is documented in `docs/operational/QUICK_START.md`.

Acknowledgments: The author acknowledges the developers and communities behind scGPT, BiomedBERT, and the Gene Expression Omnibus (GEO) for providing open-access tools and datasets that made this work possible.

Conflicts of Interest: The author declares no conflicts of interest.

Abbreviations

The following abbreviations are used in this manuscript:

AdaLN	Adaptive Layer Normalization
ALL	Acute Lymphoblastic Leukemia
ALS	Amyotrophic Lateral Sclerosis
AML	Acute Myeloid Leukemia
AMP	Automatic Mixed Precision
ARI	Adjusted Rand Index
ASD	Autism Spectrum Disorder
AUC	Area Under Curve
BCC	Basal Cell Carcinoma
BERT	Bidirectional Encoder Representations from Transformers
CE	Cross-Entropy
CFG	Classifier-free guidance
CI	Confidence Interval
CLIP	Contrastive Language-Image Pre-training
CLOP	Contrastive Language-Omics Pretraining
CRISPR	Clustered Regularly Interspaced Short Palindromic Repeats
CUDA	Compute Unified Device Architecture
DE	Differential expression
DiT	Diffusion Transformer
DivR	Diversity Ratio
EMA	Exponential Moving Average
ESC	Embryonic Stem Cell
FD	Fréchet Distance
FP16	Half-Precision Floating Point
GAN	Generative Adversarial Network
GELU	Gaussian Error Linear Unit
GEO	Gene Expression Omnibus
GPU	Graphics Processing Unit
HBV	Hepatitis B Virus
IFN	Interferon
InfoNCE	Info Noise-Contrastive Estimation
KNN	*k*-Nearest Neighbor
LinAcc	Linear Classifier Accuracy
logFC	Log Fold Change
LoRA	Low-Rank Adaptation
LPS	Lipopolysaccharide
LR	Learning Rate
MLP	Multi-Layer Perceptron
MSE	Mean Squared Error
NK	Natural Killer
NMI	Normalized Mutual Information
ODE	Ordinary Differential Equation
OOD	Out-of-Distribution
PBMC	Peripheral Blood Mononuclear Cell
PCA	Principal Component Analysis
ReLU	Rectified Linear Unit
ROC	Receiver Operating Characteristic
scGPT	Single-cell Generative Pre-trained Transformer
scRNA-seq	Single-cell RNA sequencing
SigLIP	Sigmoid Loss for Language-Image Pre-training
UMAP	Uniform Manifold Approximation and Projection
VAE	Variational Autoencoder
VRAM	Video Random Access Memory
ZCA	Zero-phase Component Analysis

Appendix A. Architecture Details

Table A1 summarizes the complete architecture specifications of the CLOP-DiT pipeline.

Table A1. Architecture specifications.

Component	Architecture	Parameters
BiomedBERT-large (frozen)	Transformer encoder	340M
CLOP text projector	MLP [1024→1024→1024→512]	1.58M
CLOP cell projector	MLP [512→512→512→512]	1.44M
DiT1D	8× AdaLN-Zero blocks, 8 heads	22.10M
scGPT encoder (frozen)	Transformer	∼51M
scGPT decoder (frozen)	Gene token prediction head	included

Appendix B. CFG Scale Sweep

Table A2 reports the full CFG sweep across 8 guidance scales with 10-step Euler integration and 69 evaluation groups. Note that these values are computed on a different subsample (69 groups × 200 cells) than the primary operating-point results in Table 2 (69 types × 200 cells with different random seeds), which accounts for minor numerical differences between the two tables.

Table A2. CFG scale sweep (10-step Euler, 69 eval groups).

CFG	KNN-1	Steering	DivR	KNN/Random
0.5	0.147	0.788	1.13	15×
1.0	0.348	0.796	0.76	35×
1.5	0.396	0.798	0.60	40×
2.0	0.407	0.806	0.53	41×
2.5	0.408	0.804	0.49	41×
3.0	0.410	0.802	0.48	41×
4.0	0.402	0.804	0.50	40×
5.0	0.380	0.798	0.55	38×

Appendix C. Training Hyperparameters

Table A3 lists the complete training hyperparameters for both CLOP alignment and DiT flow-matching stages, corresponding to `configs/clop.yaml` and `configs/dit.yaml`.

Appendix D. Supplementary Tables

Table A4. All 80 GEO datasets used in CLOP-DiT training and validation. Datasets were preprocessed to 2,000 highly variable genes per dataset. Split: Train or held-out Validation (study-level, seed 42).

#	Accession	Description	n_{cells}	Split
1	GSE103867	scRNA-seq (H. sapiens)	2,787	Val
2	GSE110686	scRNA-seq (H. sapiens)	2,992	Train
3	GSE115571	Mouse cells, LPS stimulation	442	Train
4	GSE117988	PBMCs from patients	2,638	Train
5	GSE117988	Merkel cell carcinoma tumor	2,990	Train
6	GSE120505	Aged blood cells	3,000	Train

Continued on next page

#	Accession	Description	n_{cells}	Split
7	GSE123813	Human basal cell carcinoma	3,000	Train
8	GSE123813	Human cutaneous squamous cell carcinoma	3,000	Train
9	GSE123902	Human lung adenocarcinoma	2,753	Train
10	GSE124310	Human multiple myeloma bone marrow	2,833	Train
11	GSE130148	Human fetal lung development	3,000	Train
12	GSE132509	Pediatric bone marrow mononuclear cells	2,984	Train
13	GSE138709	Human intrahepatic cholangiocarcinoma	2,891	Train
14	GSE142653	Human pituitary gland development	2,885	Train
15	GSE143423	Human leptomeningeal brain metastasis	3,000	Train
16	GSE143423	Triple-neg. breast cancer brain met.	2,332	Train
17	GSE145929	Adult mouse prostate	2,830	Train
18	GSE145929	Adult mouse urethra	2,745	Train
19	GSE148218	Human bone marrow, ALL	2,807	Train
20	GSE149655	Human early-stage lung adenocarcinoma	2,087	Train
21	GSE155109	Endothelial cells, breast cancer	3,000	Train
22	GSE155109	Stromal cells, breast cancer	3,000	Train
23	GSE163558	Human stomach cancer	2,467	Train
24	GSE165784	Human retina development	2,622	Train
25	GSE165844	Mouse LSK hematopoietic progenitors	2,881	Train
26	GSE167597	Mouse spinal cord development	3,000	Val
27	GSE168181	Human breast cancer tissue	2,863	Train
28	GSE175975	scRNA-seq (H. sapiens)	2,728	Train
29	GSE183904	Human gastric cancer	3,000	Val
30	GSE189070	Mouse astrocytes, spinal cord injury	2,960	Train
31	GSE189357	Human lung adenocarcinoma	2,886	Train
32	GSE192857	hESC differentiation time-series	2,696	Train
33	GSE212502	Human forebrain organoids	3,000	Train
34	GSE213740	Human ascending aortic wall	2,704	Val
35	GSE220913	Human HL-60 cell lines, CFTR	2,972	Train
36	GSE222002	T cells from human cancer	2,962	Train
37	GSE222369	NK cells, human lymphoma	2,778	Train
38	GSE225600	Human breast cancer tumor	2,333	Train
39	GSE225857	Colorectal cancer liver metastases	2,986	Train
40	GSE226131	Aged mouse hematopoietic stem cells	2,608	Train
41	GSE226762	scRNA-seq (H. sapiens)	830	Val
42	GSE227644	scRNA-seq (H. sapiens)	3,000	Train
43	GSE227719	Mouse KrasG12D/P53 lung organoids	2,990	Train
44	GSE228499	Human breast cancer	2,448	Train
45	GSE235787	B cells, ALL	2,798	Train
46	GSE236565	Mouse splenic CD4+ T cells	2,556	Train
47	GSE248214	scRNA-seq (H. sapiens)	2,998	Train
48	GSE253355	Human bone marrow niche	2,835	Train
49	GSE262288	Metastatic breast cancer	2,321	Train
50	GSE272187	scRNA-seq (H. sapiens)	2,639	Train
51	GSE275119	Mouse dental tissue development	2,963	Train
52	GSE277740	Mouse thalamus, SBS vs DNBSEQ	2,644	Val
53	GSE279781	scRNA-seq (H. sapiens)	2,654	Train
54	GSE280847	CD45+ leukocytes, mouse MC38 tumor	2,864	Train
55	GSE283205	Hepatoblastoma tumor tissue	2,994	Train
56	GSE283397	Human CD8+ T cells, chronic HBV	2,856	Train
57	GSE288211	Mouse pancreatic islets, Zzef1 KO	2,724	Train
58	GSE289611	Human pulmonary pleomorphic carcinoma	2,983	Train
59	GSE291166	Mouse mesenteric lymph node	2,901	Train
60	GSE299623	scRNA-seq (H. sapiens)	2,991	Train
61	GSE300862	Mouse Kras/Trp53 lung adenocarcinoma	2,716	Val
62	GSE302433	Mouse B16F10 melanoma, ZDHHC13	2,953	Val
63	GSE303309	Brain CD45+ immune cells, Alzheimer's	2,999	Train
64	GSE306567	Mouse ESCs and neural progenitors	2,454	Train
65	GSE306676	Mouse spinal cord, SOD1-G93A ALS	2,959	Train
66	GSE307261	Human B and T cells, melanoma	2,921	Train
67	GSE307774	Mouse colorectal tumors, Kras/Braf	2,505	Train
68	GSE308428	Mouse hippocampal microglia, ASD	2,991	Train
69	GSE309368	Human myeloid cells, healthy donors	1,361	Train
70	GSE315628	scRNA-seq (M. musculus)	2,895	Train
71	GSE318353	scRNA-seq (H. sapiens)	2,964	Train
72	GSE98638	Human hepatocellular carcinoma T cells	3,000	Train
73	GSE120446	Bone marrow, hematopoietic development	2,890	Train
74	GSE104323	Dentate gyrus, brain development	3,000	Train

Continued on next page

#	Accession	Description	n_{cells}	Split
75	GSE75748	Endoderm differentiation	2,531	Train
76	GSE144024	Human embryonic stem cells	3,000	Train
77	GSE72857	Hematopoiesis	3,000	Train
78	GSE226824	IFN-treated hematopoietic progenitors	2,739	Train
79	GSE141259	Lung development	3,000	Train
80	GSE139369	Human hematopoiesis (Setty et al.)	2,995	Train
	Total		**220,304**	

Table A6. Full CFG scale sweep. 200 cells generated per group (69 groups). KNN: type-identity accuracy; Steer.: alignment; DivR: diversity ratio (1.0 = ideal); LinAcc: linear classifier accuracy; FD: Fréchet distance.

CFG	Solver	Steps	KNN-1	KNN-5	Steer.	DivR	LinAcc	FD
Primary configurations (Table 2)								
3.0	Euler	10	0.367	0.554	0.802	0.473	0.508	3.46
2.0	Euler	10	0.369	0.553	0.810	0.513	0.511	3.17
1.0	Euler	10	0.313	0.483	0.810	0.745	0.390	2.84
0.0	Euler	10	0.010	0.051	0.475	1.833	0.010	0.58
3.0	Midpoint	10	0.340	0.528	0.792	0.628	0.477	3.38
2.0	Midpoint	10	0.340	0.525	0.800	0.686	0.475	3.13
1.0	Midpoint	10	0.288	0.461	0.807	0.929	0.357	2.64
3.0	Euler	50	0.348	0.545	0.795	0.622	0.489	3.33
1.0	Euler	50	0.293	0.466	0.807	0.919	0.365	2.65
—	Gaussian	—	0.011	0.048	0.466	2.277	0.009	0.03
Fine-grained CFG sweep (Euler solver)								
0.5	Euler	10	0.147	0.284	0.788	1.129	—	2.33
1.0	Euler	10	0.348	0.515	0.796	0.757	—	2.94
1.5	Euler	10	0.396	0.553	0.798	0.600	—	3.36
2.0	Euler	10	0.407	0.580	0.806	0.526	—	3.54
2.5	Euler	10	0.408	0.590	0.804	0.493	—	3.70
3.0	Euler	10	0.410	0.592	0.802	0.484	—	3.80
4.0	Euler	10	0.402	0.586	0.804	0.502	—	4.01
5.0	Euler	10	0.380	0.576	0.798	0.548	—	4.34
0.5	Euler	20	0.137	0.269	0.788	1.230	—	2.07
1.0	Euler	20	0.334	0.504	0.794	0.842	—	2.83
1.5	Euler	20	0.380	0.545	0.798	0.679	—	3.19
2.0	Euler	20	0.391	0.569	0.798	0.600	—	3.43
2.5	Euler	20	0.394	0.578	0.800	0.559	—	3.66
3.0	Euler	20	0.398	0.582	0.800	0.540	—	3.77
4.0	Euler	20	0.388	0.578	0.802	0.535	—	3.97
5.0	Euler	20	0.379	0.567	0.792	0.551	—	4.16
0.5	Euler	50	0.132	0.259	0.792	1.318	—	1.95
1.0	Euler	50	0.324	0.495	0.794	0.933	—	2.72
1.5	Euler	50	0.370	0.538	0.796	0.776	—	3.14
2.0	Euler	50	0.384	0.563	0.796	0.699	—	3.31
2.5	Euler	50	0.388	0.576	0.796	0.658	—	3.48
3.0	Euler	50	0.392	0.581	0.792	0.637	—	3.64
4.0	Euler	50	0.379	0.579	0.790	0.622	—	3.86
5.0	Euler	50	0.378	0.570	0.790	0.626	—	4.10

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Table A3. Training hyperparameters.

Hyperparameter	CLOP	DiT
Optimizer	AdamW	AdamW (β_1=0.9, β_2=0.999)
Learning rate	5×10^{-4}	2×10^{-4}
Weight decay	0.06	0.01
Batch size	1024	1024
Epochs	60	200
Warmup epochs	5	5
LR schedule	Cosine with warmup	Cosine with warmup
Gradient clipping	1.0	1.0
Mixed precision	FP16 (AMP)	FP16 (AMP)
Dropout	0.2	Proj: 0.1, Attn: 0.0
EMA decay	—	0.9999
Temperature	14.0 (fixed; see Section 2.2)	—
Loss	PrototypeSigLIP	Flow matching MSE
Cohesion weight	0.15	—
Label smoothing	0.1	—
CFG drop prob	—	0.15
Time sampling	—	Logit-normal (μ=0, σ=1)

Table A5. The 8 held-out validation datasets (study-level split, seed 42, val_split = 0.1). No cells from these studies were seen during CLOP or DiT training.

#	Accession	Description	n_{cells}
1	GSE103867	scRNA-seq (H. sapiens)	2,787
2	GSE167597	Mouse spinal cord development	3,000
3	GSE183904	Human gastric cancer	3,000
4	GSE213740	Human ascending aortic wall	2,704
5	GSE226762	scRNA-seq (H. sapiens)	830
6	GSE277740	Mouse thalamus, SBS vs DNBSEQ	2,644
7	GSE300862	Mouse Kras/Trp53 lung adenocarcinoma	2,716
8	GSE302433	Mouse B16F10 melanoma, ZDHHC13	2,953
	Total		**20,634**

Table A7. The 5 biological validation datasets used for marker gene correlation, expression distribution fidelity, and Gene Ontology enrichment analysis (Section 3.11). These datasets were selected to span diverse human cancer tissue types and immune cell populations. **Four of the five datasets are from the training split; only GSE183904 is held-out (Table S2). This analysis therefore primarily evaluates in-distribution decoder-reconstructed fidelity, not out-of-distribution generalization.**

#	GEO Accession	Tissue	Description	Split	n_{cells}
1	GSE123902	Lung	Lung adenocarcinoma TME	Train	2,753
2	GSE183904	Gastric	Gastric cancer TME	Val	3,000
3	GSE123813	Skin	Basal cell carcinoma	Train	3,000
4	GSE98638	Liver	HCC T cell landscape	Train	3,000
5	GSE132509	Blood	ALL bone marrow cells	Train	2,984
	Total				**14,737**

Table A8. Bootstrap 95% confidence intervals (percentile method, B=1,000) for headline generation metrics. Resampling unit: cell types (n=69 types resampled with replacement). Classifiers (KNN, logistic regression) were trained once on real data; only the composition of evaluated types varies across bootstrap iterations. Generation configuration: CFG = 1.5, Midpoint solver, 20 steps. Point estimates reflect the full (non-resampled) evaluation over all 69 types (100 generated cells per type, 6,900 total).

Metric	Point Est.	95% CI Lower	95% CI Upper	Boot. SE
KNN Top-1 Accuracy	0.509	0.421	0.586	0.043
KNN Top-5 Accuracy	0.728	0.658	0.794	0.035
Steering Accuracy	1.000	1.000	1.000	0.000
Diversity Ratio (DivR)	0.969	0.963	0.974	0.003
Linear Classifier Acc.	0.583	0.518	0.646	0.034
Centroid Cosine Sim.	0.898	0.875	0.913	0.009

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